

Einstein Dynamics as the Unique Minimal First-Response Metric Projection of AASC Gravity

A Kernel-Forced Same-Scope Curvature-Response Exhaustion Theorem
from Admissibility, Standing, Metric Projection, and Closed Source
Continuation

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Abstract

This manuscript derives the Einstein field-equation form as the unique minimal first-response same-scope metric projection of AASC gravity, and it does so as a standalone AASC entry point. The proof does not begin by treating a metric, curvature tensor, scalar mass, or vacuum-energy number as primitive. It first fixes the kernel roles required for any determinate same-domain theorem object: admissibility, standing, reference, and irreversibility. Those roles are not optional vocabulary. Once a metric field-equation target is used as a determinate same-scope object with fixed carrier, source role, continuation, reportability, and lawful redescription, the kernel is already active.

The argument then specializes the kernel to the physical interior. Reusable physical standing under lawful redescription forces quotient architecture. In the regular-stratum AER comparison class, quotient observables, extension-independent Poisson descent, observable separation, observable completeness, and presentation arbitrariness force the GR-type gravity role: coisotropic standing surface, characteristic redescription, local first-class constraint generators, and multiplier-type Hamiltonian presentation. Gravity in this AASC sense is therefore not primitive force, primitive curvature, primitive metric field content, or a primitive quantization target. It is the admissibility-forced constraint role of the physical interior. This manuscript includes the self-contained gravity-role derivation needed for the Einstein theorem; the corpus papers remain provenance and audit expansions, not prerequisites for reading the proof. Global functional-analytic realization, singular-stratum completion, and

boundary-domain completion are not premises of the metric normal-form theorem; they belong to realization-layer or richer-scope analysis.

The new Einstein-side contribution is the metric projection and curvature-response exhaustion. A faithful four-dimensional Lorentzian projection of the AASC gravity role licenses a minimal metric-compatible grammar (M^4, g, ∇) . A same-scope first-response metric candidate $E_{\mu\nu}$ must be internal to that licensed metric projection, tensorial under lawful redescription, rank two, symmetric, local, linear in the first admitted curvature obstruction, source-continuation-compatible, boundary-trace role-fixed, and free of hidden anchors, hidden tensor load, arbitrary selectors, curvature products, curvature derivatives, torsion, metric-affine nonmetricity, nonlocal operators, hidden nonmetric transport, boundary/global data, representative privilege, and source-side laundering. These clauses define $\text{SameScopeMetric}_1(E_{\mu\nu})$. The curvature-response exhaustion theorem proves that every such candidate has the normal form

$$E_{\mu\nu} = aR_{\mu\nu} + bRg_{\mu\nu} + cg_{\mu\nu}.$$

Closed same-scope source continuation, derived from coherent trajectory dynamics and boundary-trace role fixation, requires

$$\nabla^\mu T_{\mu\nu}^{\text{closed}} = 0,$$

and therefore requires $\nabla^\mu E_{\mu\nu} = 0$. The internal curvature identity of the licensed metric projection fixes $b = -a/2$, so $E_{\mu\nu} = aG_{\mu\nu} + cg_{\mu\nu}$. Coupling to the closed source ledger yields

$$G_{\mu\nu} + \Lambda g_{\mu\nu} = \kappa T_{\mu\nu}^{\text{closed}}.$$

The source tensor is not primitive mass-energy substance; it is the metric-facing extraction of standing-bearing deformation-response, field-carrier, particle-persistence, transport, interaction, and vacuum-baseline roles. The cosmological term is not primitive cutoff zero-point energy; it is the quotient-stable homogeneous metric-facing vacuum ledger. Numerical constants require scale and calibration anchors. High-curvature, horizon, nonlocal, torsion, metric-affine, scalar-tensor, semiclassical, and other modified-gravity structures are not denied. They are typed as skin, declared tensor load, higher-response structure, effective scope, richer scope, certificate route, open-source ledger, or standing failure unless they collapse to the minimal first-response Einstein form. The strength of the AASC proof class is precisely that it polices hidden selectors, primitive metric backflow, source laundering, and same-scope repair routes before metric dynamics is written.

Keywords: AASC; AMetric boundary; general relativity; Einstein field equations; curvature response; metric projection; source continuation; boundary trace; coherent trajectory; stress-energy; vacuum energy; cosmological constant; horizon instability; entanglement; nonmetric compatibility; scale import; admissibility; standing; fixed-domain exhaustion.

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Read This First: Proof Spine

The proof class is kernel-forced same-scope metric projection. Its correctness is determined by the declared target, hypotheses, proof links, and scope report, not by resemblance to a metric-first derivation. The proof target audits what must already be fixed for a metric field-equation candidate to stand as a determinate same-scope theorem object, then exhausts the admissible first-response metric tensors inside that target.

The proof spine is:

$$\begin{aligned} & \text{determinate same-scope metric-response target} \Rightarrow K \Rightarrow A^+ \\ & \Rightarrow \text{physical quotient architecture and AER gravity role} \\ & \Rightarrow \text{faithful four-dimensional Lorentzian metric projection} \\ & \Rightarrow \text{SameScopeMetric}_1(E_{\mu\nu}) \\ & \Rightarrow E_{\mu\nu} = aR_{\mu\nu} + bRg_{\mu\nu} + cg_{\mu\nu} \\ & \Rightarrow \nabla^\mu E_{\mu\nu} = 0 \\ & \Rightarrow E_{\mu\nu} = aG_{\mu\nu} + cg_{\mu\nu} \\ & \Rightarrow G_{\mu\nu} + \Lambda g_{\mu\nu} = \kappa T_{\mu\nu}^{\text{closed}}. \end{aligned}$$

This is not a Lovelock proof, not a phenomenological fit of GR, not a numerical derivation of κ or Λ , not a denial of declared richer-scope modified gravity, and not a quantum-gravity completion. It is a fixed-scope normal-form theorem: once the AASC gravity role is faithfully projected into minimal local Lorentzian metric language and once closed source continuation is fixed, the first-response same-scope metric equation is forced.

Correctness objections must attack named proof links. Framework familiarity is not a proof predicate: a demand that the argument first become metric-first is correctness-relevant only when it identifies a broken theorem link in this proof class.

Roadmap and exact claim

Core thesis. Einstein dynamics is the unique minimal first-response same-scope metric projection of AASC gravity. AASC first forces the gravitational role as constraint architecture. Once that role is faithfully projected into local Lorentzian metric language, once metric/source/vacuum/horizon roles are fixed by boundary-trace discipline rather than primitive representative selection, and once no extra same-scope tensor load is admitted,

the field-equation form is forced:

$$G_{\mu\nu} + \Lambda g_{\mu\nu} = \kappa T_{\mu\nu}^{\text{closed}}.$$

The result is not a derivation of all gravitational physics from empty context. It is a fixed-scope normal-form theorem with an explicit target predicate. Its proof power comes from the AASC order: target, carrier, source, continuation, reportability, and lawful redescription are fixed before metric dynamics is allowed to carry theorem force. This blocks hidden selectors, primitive metric backflow, source-side laundering, and post-hoc repair routes that a metric-first presentation can leave implicit.

The proof order is:

$$\begin{aligned} &\text{kernel target adequacy} \Rightarrow K = (\text{Adm}, \text{Stand}, \text{Ref}, \text{Irr}) \Rightarrow A^+ \\ &\Rightarrow \text{physical quotient architecture} \Rightarrow \text{AER gravity role} \\ &\Rightarrow \text{coisotropic standing, first-class closure, multiplier form} \\ &\Rightarrow \text{faithful metric projection fixed by role-anchor/boundary discipline} \\ &\Rightarrow \text{coherent metric/source trajectories and closed source continuation} \\ &\Rightarrow \text{first-response curvature exhaustion} \\ &\Rightarrow E_{\mu\nu} = aR_{\mu\nu} + bRg_{\mu\nu} + cg_{\mu\nu} \\ &\Rightarrow \nabla^\mu E_{\mu\nu} = 0 \Rightarrow E_{\mu\nu} = aG_{\mu\nu} + cg_{\mu\nu} \\ &\Rightarrow G_{\mu\nu} + \Lambda g_{\mu\nu} = \kappa T_{\mu\nu}^{\text{closed}}. \end{aligned}$$

The paper uses ATS and UEAP as governance/audit grammars. ATS types objects as anchors, tensors, or skin. UEAP fixes target, carrier, laundering exclusions, and reportable status. Neither supplies the field equation. The proof center is the metric curvature-response exhaustion theorem and its coupling to the closed source ledger. Boundary-trace fixation and coherent trajectory dynamics ground closed source continuation. The vacuum/ZPE module types $\Lambda g_{\mu\nu}$ as homogeneous quotient-stable ledger content. The entanglement/nonmetric-compatibility module blocks hidden boundary communication. The high-curvature/horizon module closes endpoint repair routes. These support the declared scope while leaving SameScopeMetric₁ exhaustion as the central proof.

Part I

Standalone AASC Foundation and Proof-Class Lock

1 Proof-class lock and correctness-relevant criticism

The proof class is kernel-forced AASC metric-projection normal form. It is evaluated by its named proof links. A metric-first derivation may be valuable as a comparator or an AASC-free reconstruction route, but it is not the correctness standard for this theorem unless it identifies a failure in the fixed AASC proof chain.

Definition 1.1 (Correctness-relevant criticism). *A criticism is correctness-relevant to this manuscript only if it identifies a false premise, invalid inference, hidden import, scope shift, overreported conclusion, or broken named link in the proof chain: target adequacy to the kernel; AMetric boundary discipline; physical quotient architecture; AER gravity-role forcing; faithful metric projection; boundary-trace role fixation; coherent source continuation; the predicate SameScopeMetric₁; the curvature-response exhaustion; divergence compatibility; stress-energy extraction; vacuum quotienting; high-curvature/horizon route closure; modified-gravity classification; or reportable-scope discipline.*

Remark 1.2 (Proof-class mismatch). *A demand that the proof be rewritten as a Lovelock theorem, an Einstein-Hilbert action variation, a phenomenological GR derivation, or a quantum-gravity program changes the proof class unless it identifies a broken AASC theorem link. AASC is not an overlay placed on top of conventional physics. It is the fixed-domain governance discipline for theorem objects before a metric presentation is licensed.*

Proposition 1.3 (Proof-class boundary). *Within the declared proof class, the relevant correctness question is whether the metric-response target, kernel roles, gravity role, metric projection, first-response predicate, source closure, and divergence calculation are valid. Framework familiarity or preference for a standard derivation has no bearing on the theorem unless it locates a mathematical or role-typing failure in those links.*

Proof. A proof object is evaluated relative to its target, carrier, hypotheses, inference links, and reportable scope. Framework preference changes none of these. If an objection changes the target to a different proof class, it asks for a different theorem. If it preserves the target and identifies a broken link, it is correctness-relevant. These alternatives exhaust relevance for this manuscript. $\square$

2 Kernel-neutral target adequacy and the AASC kernel

This section makes the kernel use self-contained. The kernel is not assumed as doctrine; it is the role structure forced by determinate same-domain construction. The Einstein theorem uses this structure because a same-scope metric field-equation candidate is not raw syntax. It is a theorem-bearing object with target, carrier, source role, continuation, act-time status, and lawful redescription.

Definition 2.1 (Kernel-neutral target adequacy). *A theorem-bearing act is target-adequate when it satisfies the following four conditions.*

1. **Target determinacy:** *the act has a repeatable object of evaluation, not a presentation-dependent trace.*
2. **Step/status evaluability:** *its assertions and transitions are evaluable as advancing, blocking, or failing relative to the same target.*
3. **Act-time finality:** *later repair or reclassification is a distinct act, not retrospective standing for the same act.*
4. **Same-regime fidelity:** *lawful redescription preserves target, carrier, role, standing status, and continuation profile.*

Definition 2.2 (AASC kernel). *The AASC kernel is*

$$K = (\text{Adm}, \text{Stand}, \text{Ref}, \text{Irr}),$$

where *Adm* distinguishes licensed from unlicensed use, *Stand* is reusable theorem-bearing act-status, *Ref* fixes the same-domain target, and *Irr* blocks same-act post-hoc repair of a failed act.

Kernel role	Governance work
Reference	Fixes target, carrier, object identity, and lawful redescription.
Standing	Supplies theorem-bearing or counterexample-bearing force and licensed reuse.
Admissibility	Fixes the boundary between licensed use, support, bookkeeping, repair, scope change, selector import, and reportability.
Irreversibility	Blocks same-act repair, deferred certification, retrospective reclassification, and post-hoc standing creation.

Theorem 2.3 (Target adequacy forces the kernel). *If a theorem-bearing act is target-adequate, then the kernel $K = (\text{Adm}, \text{Stand}, \text{Ref}, \text{Irr})$ is already in force for that act.*

Proof. Target determinacy requires a role that preserves a fixed object across use and redescription; that role is reference. Step/status evaluability requires a role by which an assertion can carry licensed force for later use; that role is standing. Act-time finality requires a role preventing later intervention from changing the status of the same act; that role is irreversibility. Same-regime fidelity requires a boundary under which the previous three roles remain jointly enforceable; that boundary is admissibility. Thus the kernel is not optional terminology added after the fact. It is the minimal governance work already performed by any target-adequate theorem-bearing act. $\square$

Corollary 2.4 (Vocabulary independence). *A critic may reject the names Adm, Stand, Ref, Irr, but if the critic preserves determinate target identity, evaluable theorem status, act-time distinction between continuation and repair, and same-regime fidelity, the critic preserves the kernel roles under other names. If any of those roles is denied, the same theorem-bearing metric-response target is no longer preserved.*

Lemma 2.5 (Reference from target determinacy). *Target determinacy forces reference. If no role fixes the repeatable object of evaluation, then redescription can change what is being evaluated while preserving the same marks. The result is a trace or presentation change, not the same theorem-bearing act.*

Proof. A theorem-bearing act must retain a determinate target through reuse, comparison, and lawful redescription. Without such fixation, a later use cannot be distinguished from replacement, relabeling, or change of object. The role that preserves target identity under those operations is reference. Thus target determinacy is not a label for reference; it is the burden that forces the reference role. $\square$

Lemma 2.6 (Standing from step/status evaluability). *Step/status evaluability forces standing. If an assertion can be evaluated as advancing, blocking, or failing relative to a target, then it must have an act-status that determines whether it may be used in licensed continuation.*

Proof. A raw expression may occur without theorem force. To function inside a proof, it must be eligible or ineligible for downstream use relative to the fixed target. That eligibility is standing. If no such role exists, the expression remains inscription, support, or bookkeeping rather than a theorem-bearing step. $\square$

Lemma 2.7 (Irreversibility from act-time finality). *Act-time finality forces irreversibility. If a later correction can convert a failed act into the same admissible act, then failure and success were not determinate statuses of the original act.*

Proof. A corrected later performance may stand as a new act. It cannot make the earlier failed performance the same admissible act without erasing the distinction between continuation and repair. The governance role that blocks this erasure is irreversibility. $\square$

Lemma 2.8 (Admissibility as joint enforceability boundary). *Same-regime fidelity forces admissibility. A fixed theorem domain must preserve target identity, standing status, act-time finality, and continuation profile under lawful redescription. The boundary under which those classifications remain jointly enforceable is admissibility.*

Proof. If lawful redescription can change which acts are licensed, which targets are fixed, or which failures can be repaired, then the regime has no stable theorem object. If those classifications are preserved, there is already a boundary separating licensed from unlicensed use and continuation from repair. That boundary is admissibility. $\square$

Proposition 2.9 (Anti-weakening burden). *A proposed replacement for the kernel is correctness-relevant only if it preserves the same target, theorem-bearing force, act-time finality, lawful redescription, continuation profile, and reportability while omitting one of the kernel roles and without performing the same governance work under another name.*

Proof. If the replacement loses one of the preservation conditions, it changes the target rather than weakening the kernel. If it restores the lost condition by another selector, classifier, repair channel, or status rule, it has reintroduced the omitted governance work. If it changes no theorem-bearing use, it is bookkeeping. These cases exhaust same-domain weakening attempts. $\square$

3 Fixed-domain consequence layer A^+

The Einstein proof uses a compact consequence package of the kernel. Write

$$\text{nondegenerate determinate same-domain use} \Rightarrow K \Rightarrow A^+.$$

The package A^+ contains only consequences needed for this theorem.

Definition 3.1 (AASC consequence layer for metric projection). *For a fixed theorem domain, A^+ consists of the following consequences.*

1. **AMetric boundary:** *no metric, ranking, grading, probability, selector, canonical representative, comparison scale, or boundary parameter has primitive admissibility authority.*

2. **No second same-scope classifier:** a status factor that changes theorem standing is anchor motion, declared tensor load, or forbidden hidden selector work; if it changes no standing, it is skin/bookkeeping.
3. **No same-act repair:** later correction, deferred coherence, or retrospective reclassification is a new act or scope change.
4. **Standing as reuse-stable admissibility:** standing is admissibility preserved under licensed same-domain continuation.
5. **Unique admissible interior:** plural same-domain admissible interiors collapse extensionally or introduce a selector.
6. **Report preservation:** a report may not claim more theorem force than the act carries.
7. **No carrier/scope transfer without declaration:** moving dimension, signature, connection, source ledger, boundary data, locality, or field content changes scope unless declared as an anchor or tensor load.

Theorem 3.2 (No independent same-scope status discriminator). *In a fixed nondegenerate theorem domain, no independent same-scope status discriminator can carry theorem force. Any proposed discriminator is one of: governance-equivalent to the admitted interface, bookkeeping/skin, declared anchor motion, declared tensor load, or forbidden hidden selector work.*

Proof. If the discriminator changes standing-bearing use, it performs admissibility-relevant work. In a fixed domain, such work is already governed by Adm, Stand, Ref, and Irr. If the discriminator preserves those roles exactly, it is governance-equivalent. If it changes target, carrier, source, connection, boundary, or continuation, it moves anchors or adds tensor load. If it changes no standing-bearing use, it is skin. The remaining possibility is undeclared same-scope authority, which violates the kernel boundary and is forbidden. $\square$

4 Dependency-inversion exclusion for Einstein dynamics

The dependency order is not

optional AASC framework $\Rightarrow$ Einstein equation.

The dependency order is

$$\begin{aligned} &\text{determinate same-scope metric field-equation target} \Rightarrow K \Rightarrow A^+ \\ &\Rightarrow \text{metric-response admissibility discipline.} \end{aligned}$$

AASC names and audits the forced roles. It is not a philosophical supplement attached to an already well-typed metric equation.

Definition 4.1 (Metric-response theorem act). *A metric-response theorem act is a use of an expression $E_{\mu\nu} = \kappa S_{\mu\nu}$ as a standing-bearing same-scope field-equation candidate in a fixed metric projection. Such an act fixes the metric carrier, tensor type, source role, continuation rule, redescription behavior, and reportable scope.*

Theorem 4.2 (Metric-response dependency inversion). *Every nondegenerate same-scope metric-response theorem act already instantiates the kernel roles. Therefore AASC discipline is not an external condition imposed on metric dynamics; it is the governance structure required for the metric-response target to be determinate.*

Proof. A same-scope metric-response theorem act must fix what tensor is being evaluated, on which carrier, under which redescription law, with which source ledger, under which continuation rule, and with what reportable status. These are precisely target determinacy, step/status evaluability, act-time finality, and same-regime fidelity. By the target-adequacy theorem, the kernel is active. Hence the AASC discipline is already required before the metric equation can stand as a theorem object. $\square$

5 Cost of kernel denial in the Einstein theorem

Kernel denial does not preserve the same Einstein theorem with fewer assumptions. It changes or destroys the determinate metric-response object.

Kernel role denied	Einstein-specific cost
Reference	No fixed metric-projection target; no stable meaning for (M, g, ∇) , $E_{\mu\nu}$, $T_{\mu\nu}^{\text{closed}}$, source closure, dimension, signature, or same-scope comparison class.
Standing	The equation remains syntax, modeling preference, or comparator language, but lacks theorem-bearing metric-response force and licensed downstream use.

Kernel role denied	Einstein-specific cost
Admissibility	No boundary separates same-scope first response from richer-scope tensor load, support, bookkeeping, source laundering, vacuum scalar import, representative privilege, hidden selector, or undeclared boundary repair.
Irreversibility	Deferred coherence, later source repair, retrospective vacuum reinterpretation, post-hoc horizon certificate, and high-curvature same-act repair reopen.

Corollary 5.1 (No third position). *There is no position from which one may reject the kernel while still using $E_{\mu\nu} = \kappa T_{\mu\nu}$ as a determinate same-scope metric field-equation theorem object. Either the metric-response target remains nondegenerate, in which case the kernel is active, or one of the costs in the table is paid and the same theorem object is lost.*

Part II

AASC Gravity Before Metric Dynamics

6 Admissible physical interior and quotient architecture

Definition 6.1 (Admissible physical interior). *An admissible physical interior is a domain supporting standing-bearing physical contents whose lawful continuation preserves same-domain identity, whose redescription preserve quotient identity, whose observables carry reusable physical standing, and whose failed uses cannot be retroactively converted into standing without repair or scope change.*

Definition 6.2 (Physical presentation and quotient). *A physical presentation consists of a state carrier P , outcome-relevant observables $\mathcal{O}$, a lawful redescription relation $\sim$, and an admissible continuation class Cont . Its physical content is the quotient $P/\sim$ whenever that quotient exists in the required regularity class.*

Theorem 6.3 (Quotient necessity for physical standing). *If a physical domain supports*

reusable same-domain physical content under lawful redescription, then its standing-bearing content factors through a quotient by admissible redescription.

Proof. If two lawfully redescribed presentations are not identified in physical standing, the theory privileges a representative or imports a hidden discriminator. Representative privilege violates reference stability, and hidden discrimination violates admissibility because it supplies unwitnessed same-domain status. Therefore lawfully redescribed states must be identified. Conversely, the quotient retains exactly those distinctions that survive admissible witnesses and continuation. $\square$

Definition 6.4 (Admissibility-bearing quotient architecture). *An admissibility-bearing quotient architecture is*

$$A = (\text{Anc}_A, \text{Ten}_A, \text{Skin}_A, \sim_A, \text{Cont}_A),$$

where anchors fix domain and admissibility envelope, tensors carry invariant standing-bearing content, skin is representational surplus, $\sim_A$ is lawful identity-preserving equivalence, and Cont_A is admissible continuation.

Theorem 6.5 (No fourth same-domain role). *Within a fixed admissibility-bearing physical presentation, every descriptive element is anchor, tensor, or skin. No fourth same-domain standing-bearing role survives.*

Proof. If varying an element changes the domain, boundary, regime, or well-typedness conditions, it is anchor. With anchors fixed, if varying it changes invariant physical content, observable witness, compatibility relation, or continuation output, it is tensor. If it changes neither, no admissible downstream use distinguishes it; it is skin. A fourth role would carry standing-bearing force while changing neither domain nor invariant content, which is hidden selector work or bookkeeping. Both are excluded. $\square$

7 Regular-stratum AER specialization

The AER comparison class is the local Hamiltonian observable-evaluation regime in which the AASC physical interior is analyzed before metric projection. It is not GR smuggled into the premises. It fixes the minimal regular setting in which quotient observables, redescription, and continuation can be evaluated by Hamiltonian geometry.

Let (S, ω) be a local symplectic state carrier, let $C \subset S$ be a smooth standing surface, let $\pi : C \rightarrow S_{\text{phys}}$ be a smooth quotient submersion, and define the redescription distribution

$$D = \ker(d\pi).$$

Let ω_C be the pullback of ω to C . Physical observables are smooth functions on C constant on π -fibres.

Definition 7.1 (AER interface). *The regular-stratum AER interface consists of the following clauses.*

1. **AER-Obmin:** *Poisson brackets of physical observables descend to the quotient extension-independently.*
2. **AER-Sep:** *physical observables separate quotient-transverse directions and their differentials span $\text{Ann}(D_x)$.*
3. **AER-Max:** *no nonzero presymplectic same-scope direction remains invisible to all physical observables.*
4. **AER-P:** *presentation-parametrized evolution induces the same reduced physical continuation up to redescription directions and dynamically inert reduced content.*

Lemma 7.2 (Characteristic capture). *Under AER-Obmin and AER-Sep,*

$$(T_x C)^{\perp_\omega} \subseteq D_x.$$

Consequently C is coisotropic.

Proof. A Hamiltonian generated by a function vanishing on C has bracket zero with every physical observable on C by extension-independent descent. Quotient separation then forces its Hamiltonian direction to annihilate all quotient-observable differentials and hence to lie in the vertical redescription distribution. This gives characteristic capture and coisotropy. $\square$

Lemma 7.3 (No hidden presymplectic remainder). *Under AER-Max,*

$$D_x \subseteq \ker(\omega_{C,x}).$$

Proof. If $v \in D_x$ were not in $\ker(\omega_{C,x})$, AER-Max would supply a physical observable detecting v . But physical observables are constant on π -fibres, so they annihilate D_x . Contradiction. $\square$

Theorem 7.4 (Local AER forcing theorem). *Under AER-Obmin, AER-Sep, and AER-Max,*

$$D = \ker(d\pi) = \ker(\omega_C).$$

Thus the standing surface is coisotropic and admissible redescription coincides with the characteristic foliation of ω_C . Local coisotropic normal form supplies first-class constraint generators, and AER-P yields multiplier-type Hamiltonian presentation.

Proof. Characteristic capture gives $\ker(\omega_C) = (TC)^\perp \subseteq D$. AER-Max gives $D \subseteq \ker(\omega_C)$. Hence equality holds. Coisotropic normal form provides local defining constraints ϕ_a spanning D . Since their Hamiltonian flows are tangent to the standing surface, their Poisson brackets vanish on C and are therefore first-class locally. Presentation-parametrized Hamiltonians differ from reduced physical continuation by combinations of these constraints, giving multiplier form. $\square$

Proposition 7.5 (AER-Max minimality witness). *AER-Max is necessary rather than decorative. There are regular local models satisfying quotient descent and observable separation while retaining a hidden presymplectic same-scope direction unless AER-Max is imposed.*

Proof. Let $S = \mathbb{R}^4_{(q_1, p_1, q_2, p_2)}$ with $\omega = dq_1 \wedge dp_1 + dq_2 \wedge dp_2$, and let $C = \{p_1 = 0\}$. Then C is coisotropic and $\ker(\omega_C) = \text{span}\{\partial_{q_1}\}$. Define $\pi : C \rightarrow \mathbb{R}$ by $\pi(q_1, q_2, p_2) = q_2$. Then $D = \ker(d\pi) = \text{span}\{\partial_{q_1}, \partial_{p_2}\}$, so $\ker(\omega_C) \subsetneq D$. Physical observables are functions of q_2 ; their brackets descend and their differentials span $\text{Ann}(D)$, but every such observable annihilates ∂_{p_2} , even though $\partial_{p_2} \notin \ker(\omega_C)$. Thus descent and separation alone leave an invisible presymplectic remainder. AER-Max is the observable-completeness condition that excludes this undeclared same-scope direction. $\square$

Remark 7.6 (Local regular-stratum and field-theoretic scope). *The AER proof is a local regular-stratum structural normal-form argument. It is not a global functional-analytic theorem for every singular stratum, boundary condition, or infinite-dimensional completion of the ADM phase space. Its role here is to fix the gravity-role architecture: coisotropic standing, characteristic redescription, local first-class generators, and multiplier presentation. Global analytic completions, boundary domains, and singular strata are realization-layer or richer-scope matters, not hidden premises in the Einstein-form exhaustion.*

8 AASC gravity necessity

Definition 8.1 (AASC gravity). *Gravity in the AASC sense is the structural role corresponding to the constraint architecture required for admissible physical construction: standing surface, admissible redescription, observable quotient, constraint closure, and multiplier-type Hamiltonian presentation.*

Definition 8.2 (GR-type canonical constraint architecture). *A theorem object has GR-type canonical constraint architecture when its standing-bearing interior contains, up to lawful role-preserving equivalence:*

1. a standing surface C in a Hamiltonian state carrier;

2. *physical observables factoring through a quotient $\pi : C \rightarrow S_{\text{phys}}$;*
3. *admissible redescription coinciding with the characteristic foliation of ω_C ;*
4. *local first-class constraint generators spanning admissible redescription;*
5. *presentation-parametrized dynamics in multiplier form.*

Theorem 8.3 (AASC gravity necessity). *If a system instantiates an admissible physical interior in the regular-stratum AER comparison class and satisfies presentation arbitrariness, then it necessarily instantiates gravity in the AASC sense: a GR-type canonical constraint architecture.*

Proof. The admissible physical interior supplies standing, reference, lawful redescription, quotient identity, and continuation. Quotient necessity gives physical observables factoring through $\pi : C \rightarrow S_{\text{phys}}$. The local AER forcing theorem identifies admissible redescription with the characteristic foliation of ω_C , making C coisotropic. Coisotropic normal form gives local constraint generators; regularity gives first-class closure; presentation arbitrariness gives multiplier-type Hamiltonian presentation. These are exactly the clauses of AASC gravity. $\square$

Remark 8.4 (What this gravity theorem does and does not do). *The theorem fixes the gravitational role before metric dynamics. It does not derive the Einstein field equations, does not make curvature primitive, and does not forbid richer-scope gravitational theories. It supplies the constraint role that the rest of this paper faithfully projects into metric language and then exhausts at first response.*

Part III

Metric Projection and Einstein-Form Exhaustion

9 AMetric nonmetricity, metric projection, and end-point discipline

AASC uses nonmetricity in its foundational sense: the AMetric boundary is prior to every admitted metric role. In this paper AMetricity is the no-selector boundary for metric projection, not a decorative metaphysical absence claim. This must be distinguished from

metric-affine nonmetricity, which is a post-metric geometric tensor condition inside an already admitted connection theory.

Definition 9.1 (AMetric nonmetricity). *AMetric nonmetricity is the pre-admissible non-selection condition. At the AMetric boundary no metric, distance, ordering, probability, quotient, representative, selector, scale, comparison relation, source ledger, or continuation authority has standing. It is not a zero-distance metric and not a metric-affine geometry.*

Definition 9.2 (Metric-affine nonmetricity). *Metric-affine nonmetricity is a post-admission geometric structure inside an already admitted metric-affine presentation:*

$$Q_{\lambda\mu\nu} := \nabla_\lambda g_{\mu\nu} \neq 0.$$

It is a connection/comparison tensor load in a richer geometric scope, not the AMetric boundary.

Theorem 9.3 (AMetric-to-metric projection order). *No metric expression, curvature tensor, field equation, distance relation, causal order, or comparison scale has primitive authority at the AMetric boundary. Metricity becomes admissible only as a role-fixed projection after standing, reference, quotient identity, and continuation are licensed.*

Proof. A metric expression compares, orders, or relates candidates. At the AMetric boundary no comparison, ordering, representative, scale, or selector is admitted. If a metric expression is used before admission, it is a hidden selector. If it is used after admission, it is no longer boundary primitive; it is an admitted anchor, tensor, or skin element. Therefore metricity is downstream of admissibility. Curvature and field equations are still later because they presuppose the metric projection. $\square$

Theorem 9.4 (Scale import and numerical endpoint discipline). *A dimensional numerical value has AASC standing only relative to an admitted scale anchor, calibration ledger, or comparison carrier. Therefore the present theorem derives field-equation form and source/vacuum role typing; it does not attempt to generate absolute dimensional values for κ , Λ , or matter-sector scales from the AMetric boundary.*

Proof. Dimensional values compare a quantity against a unit, standard, or calibration procedure. At the AMetric boundary no scale or comparison carrier has standing. Once a scale anchor is admitted, dimensional values may stand relative to that anchor. Thus numerical constants are not boundary-generated outputs in this theorem; they are different endpoint types requiring scale import. $\square$

We use the guiding distinction

form standing $\neq$ absolute scale standing.

The theorem derives the standing of the Einstein-form projection. The numerical values of κ and Λ require scale/cosmological calibration ledgers.

10 ATS and UEAP control of the Einstein projection

The proof uses two control grammars. The anchor/tensor/skin decomposition controls the objects that appear in the theorem. UEAP controls the status of the theorem as a reportable claim. They are not additional proof engines for the Einstein-form result; they prevent role confusion and claim over-promotion. ATS types the objects, UEAP fixes the claim status, and SameScopeMetric₁ carries the field-equation proof.

Definition 10.1 (ATS role typing). *In a fixed admissibility-governed domain, every admissibility-bearing descriptive element is typed as one of three roles. An anchor is an admissibility-defining precondition. A tensor is quotient-level invariant or compatibility content carrying standing-bearing force. Skin is representational surplus with no admissibility-bearing authority. No fourth same-domain role is available without scope change, hidden tensor load, or collapse to bookkeeping.*

Proposition 10.2 (ATS ledger for the Einstein projection). *The objects used in the Einstein projection theorem have the following role ledger.*

<i>Role</i>	<i>Einstein-projection content</i>
<i>Anchors</i>	<i>AMetric boundary discipline; admissibility and standing; the faithful four-dimensional Lorentzian projection envelope; metric-compatibility in the minimal projection; closed-source scope; boundary-trace role fixation; scale/calibration anchors for dimensional quantities.</i>
<i>Tensors</i>	<i>Ricci first-response content $R_{\mu\nu}$; scalar-trace return $Rg_{\mu\nu}$; Einstein combination $G_{\mu\nu}$; closed source ledger $T_{\mu\nu}^{\text{closed}}$; homogeneous vacuum ledger $\Lambda g_{\mu\nu}$; source deformation-response; horizon certificate content when declared.</i>
<i>Skin</i>	<i>Coordinates, frames, chart labels, gauge representatives, notation, diagrams, table layout, convention choices that preserve tensor content, and metric uniqueness comparison language when used only to orient the reader.</i>

Proof. The listed anchors fix the scope in which the question is well typed. The listed tensors change standing-bearing metric, source, vacuum, or boundary content when varied.

The listed skin elements can be changed without changing standing-bearing content. This is the fixed-domain ATS partition. A proposed fourth role either changes anchors, changes tensor content, or changes no admissibility-bearing content. $\square$

Corollary 10.3 (ATS reading of SameScopeMetric₁). *The predicate SameScopeMetric₁(E_{μν}) is the metric-side ATS discipline in formal form. It forbids anchor motion, undeclared tensor load, and skin carrying force. Higher curvature, torsion, metric-affine nonmetricity, nonlocality, and extra fields are not declared false; they are excluded only from the minimal first-response same-scope class unless retyped as declared anchors or tensor load.*

Definition 10.4 (UEAP status of the main claim). *Let c_E be the claim that*

$$G_{\mu\nu} + \Lambda g_{\mu\nu} = \kappa T_{\mu\nu}^{\text{closed}}$$

is the unique minimal first-response same-scope metric projection of AASC gravity. UEAP audits c_E by fixing its target, carrier, admissible alternatives modulo skin, laundering routes, and reportable status.

Proposition 10.5 (UEAP audit of the Einstein-projection claim). *The claim c_E has the following UEAP status ledger.*

<i>UEAP coordinate</i>	<i>Value in this manuscript</i>
<i>Target</i>	<i>The minimal first-response same-scope metric field-equation form of AASC gravity.</i>
<i>Carrier</i>	<i>In-manuscript AASC gravity-role derivation; AMetric non-selection; ATS role typing; boundary-trace and coherent-trajectory support; SameScopeMetric₁ predicate; curvature-response normal-form proof; closed source-ledger extraction; vacuum/ZPE quotient discipline.</i>
<i>Scope</i>	<i>Faithful four-dimensional Lorentzian metric projection with dimension and signature treated as projection anchors unless separately derived.</i>
<i>Admissible alternatives</i>	<i>Curvature products, curvature derivatives, torsion, metric-affine nonmetricity, nonlocality, scalar/vector/tensor extensions, semiclassical mixed sources, and horizon endpoint responses, each classified as skin, declared tensor load, higher response, effective scope, richer scope, certificate route, or standing failure.</i>

<i>UEAP coordinate</i>	<i>Value in this manuscript</i>
<i>No-laundering discipline</i>	<i>The proof does not launder external metric-uniqueness comparison into proof authority, does not launder zero-point vacuum sums into Λ, does not launder numerical scale into boundary generation, and does not launder hidden high-curvature repair into same-scope dynamics.</i>
<i>Reportable status</i>	<i>Legitimate theorem within the declared minimal first-response four-dimensional Lorentzian metric-projection scope.</i>
<i>Non-reportable promotions</i>	<i>All gravitational dynamics; numerical prediction of κ or Λ; all matter Lagrangians; exclusion of all modified gravity; complete quantum gravity; complete horizon endpoint dynamics.</i>

The reportable formulation is therefore:

Within a faithful four-dimensional Lorentzian metric projection of AASC gravity, and under the minimal first-response same-scope predicate SameScopeMetric_1 , the unique continuation-compatible metric field-equation form is

$$G_{\mu\nu} + \Lambda g_{\mu\nu} = \kappa T_{\mu\nu}^{\text{closed}}.$$

This UEAP status is stronger than compatibility and weaker than a claim to all gravitational dynamics. It is a final status for the declared target class.

11 From AASC gravity role to metric field-equation form

The standard form of general relativity begins after metric gravity is already in place. A Lorentzian metric is given, its curvature is computed, and source content is represented by a stress-energy tensor. The familiar field equation is

$$G_{\mu\nu} + \Lambda g_{\mu\nu} = \kappa T_{\mu\nu}.$$

That formulation is physically successful, but it begins at the level of a geometric presentation. AASC asks what has to be true before that presentation is licensed.

AASC does not first identify gravity with force, curvature, metric field content, or a quantization target. It identifies gravity as the constraint role required for an admissible physical interior: standing surface, quotient observables, admissible redescription, first-class closure, and multiplier-type Hamiltonian presentation. The preceding sections derive this GR-type role in the form needed for the present theorem. This section begins the metric field-equation layer.

The central question is narrow:

Once the AASC gravity role is faithfully projected into local Lorentzian metric language, once the metric/source roles are fixed by admissibility-preserving boundary trace rather than selected representatives, and once no extra same-scope tensor load is admitted, what metric source equation remains possible?

The answer is Einstein form. The proof is an AASC exhaustion argument. A same-scope metric field equation must use standing-bearing metric tensor content, preserve lawful redescription, couple to an admissible closed source ledger, preserve coherent source continuation, reject primitive vacuum scalar import, and block hidden horizon repair. When those restrictions are enforced, the geometric response side collapses to three primitive slots:

$$g_{\mu\nu}, \quad R_{\mu\nu}, \quad Rg_{\mu\nu}.$$

The continuation-compatible combination is uniquely Einstein:

$$R_{\mu\nu} - \frac{1}{2}Rg_{\mu\nu}.$$

11.1 Why the proof must be explicit

The paper has one especially load-bearing step: the first-response curvature exhaustion theorem. It is not a preference for simple equations. It is a same-scope AASC normal-form theorem with explicit type restrictions and case exclusions. Higher-curvature, torsion, metric-affine nonmetricity, scalar-tensor, nonlocal, semiclassical, and horizon-endpoint structures are not declared false. They are typed. They are richer-scope, higher-response, effective, mixed-projection, or certificate-bearing structures unless they collapse to the minimal Einstein form.

The second load-bearing step is source continuation. In a short argument one might simply write $\nabla^\mu T_{\mu\nu} = 0$. This paper does not treat that equation as a primitive conservation postulate. It derives the closed-source condition from the AASC trajectory theory: a coherent same-scope trajectory is a legal trajectory whose states preserve standing, and

no incoherent prefix can be repaired by later continuation. The metric projection of a closed source trajectory is therefore divergence-compatible. If it is not, some source ledger is incomplete, an exchange channel is omitted, an anomaly/failure is present, or the scope is open/effective.

11.2 Claim-level separation

The paper separates five levels:

1. **AASC kernel layer:** $K = (\text{Adm}, \text{Stand}, \text{Ref}, \text{Irr})$.
2. **Gravity role layer:** AASC/AER forces GR-type constraint architecture.
3. **Metric projection layer:** a faithful Lorentzian metric projection represents the gravity role geometrically.
4. **Trajectory and boundary-trace layer:** legal dynamics becomes coherent standing-preserving continuation, while metric/source/vacuum/horizon roles are fixed as role-anchor classes without representative selection.
5. **Field-equation form layer:** first-response exhaustion and source continuation force Einstein form.
6. **Numerical calibration layer:** the values of κ , Λ , and matter-sector parameters require further anchors.

The new proof contribution is the metric curvature-response normal form and its coupling to the closed source ledger. The kernel, quotient, and gravity-role chain is established in the self-contained proof chain above; the corpus papers supply fuller audit expansions. Numerical calibration remains outside the endpoint of this theorem.

The surrounding AASC corpus supplies extended proofs and specialized elaborations of AMetric non-selection, metric licensing, boundary-trace role fixation, coherent trajectory dynamics, mass/inertia role structure, vacuum/ZPE quotient discipline, entanglement-style nonmetric compatibility, and horizon route closure. The present manuscript contains the amount needed to audit the Einstein theorem itself.

12 Provenance ledger: expanded here, cited for full audit

This manuscript includes the kernel, quotient, AER gravity-role, and metric-projection machinery needed to read the Einstein theorem as a standalone AASC result. The

surrounding corpus remains cited for fuller proofs, historical dependency lineage, and audit expansion. The main text uses the following provenance ledger; these items are not prerequisites for understanding the proof center.

Provenance layer	Function in this paper
AASC gravity theorem	Established in the self-contained proof chain above; supplies the forced GR-type constraint role: standing surface, observable quotient, admissible redescription, first-class closure, and multiplier form.
AMetric and metric licensing	Expanded and applied here; blocks primitive metricity and fixes metric projection as downstream from admissibility, standing, and quotient identity.
ATS role decomposition	Used in-main; types theorem objects as anchors, tensors, or skin; no fourth same-scope role is available.
Boundary trace and coherent trajectory	Stated in-main at theorem-use level; fix metric/source/vacuum roles without representative selection and ground closed source continuation.
Mass and inertia arc	Supplies deformation-response as the source-side basis, rather than primitive scalar mass.
Vacuum and ZPE arcs	Supply non-null vacuum baseline and quotient discipline for the homogeneous $\Lambda g_{\mu\nu}$ ledger.
Entanglement and non-metric compatibility	Blocks hidden nonmetric communication or boundary transport as a repair route.
Horizon instability arc	Closes high-curvature and horizon endpoint repair routes by certificate, leakage, non-sink quotient, or richer scope.
UEAP	Fixes the theorem claim's target, carrier, laundering exclusions, and reportable status.
Present paper	Proves SameScopeMetric ₁ ($E_{\mu\nu}$), the curvature-response normal form, and the closed source coupling bridge.

13 Projection architecture

Definition 13.1 (Admissibility-bearing quotient architecture). *An admissibility-bearing quotient architecture is a role-labelled object*

$$A = (\text{Anc}_A, \text{Ten}_A, \text{Skin}_A, \sim_A, \text{Cont}_A),$$

where anchors fix domain, boundary, regime, and admissibility envelope; tensors carry standing-bearing content; skin is representational surplus; $\sim_A$ is lawful identity-preserving equivalence; and Cont_A is admissible continuation.

Definition 13.2 (Projection). *A projection of A to a theorem object X is a role-preserving map*

$$\Pi_X : A \rightarrow X$$

whose kernel consists only of skin-level redundancy relative to the fixed comparison class. Projection is not conversion from one formalism to another. It is quotient-realization of role content in a target presentation.

Definition 13.3 (Gravitational projection). *A gravitational projection is the image of the AASC gravity role in a theorem object supporting constrained-Hamiltonian architecture. It presents standing surface, quotient observables, characteristic redescription, first-class closure, and multiplier dynamics in a role-specific form.*

Definition 13.4 (Metric projection). *A metric projection is a further projection*

$$\Pi_g : A_G \rightarrow (M, g_{\mu\nu})$$

that represents quotient comparison, transport compatibility, local continuation, and redescription structure geometrically.

Proposition 13.5 (No formalism-to-formalism primacy). *The metric field equation, if derived, is a theorem object of the metric projection. It is not the primitive generator of the AASC gravity role.*

Proof. The gravity role is already fixed by admissibility, standing, quotient, redescription, and continuation requirements. If the metric equation were used to authorize those requirements, it would retroactively supply boundary standing from inside the projection. That is metric backflow. AASC boundary discipline forbids using a downstream presentation to authorize the standing of the architecture that licensed the presentation. Therefore the field equation can only be a projected theorem object, not a primitive generator. $\square$

Part IV

Faithful Metric Projection

14 Metric projection licensing

Definition 14.1 (Faithful metric projection). *A metric projection $\Pi_g : A_G \rightarrow (M, g_{\mu\nu})$ is faithful iff:*

1. *the metric is downstream from the AASC gravity role;*
2. *every standing-bearing metric distinction is anchor or tensor content;*
3. *every non-standing metric distinction is skin;*
4. *no metric fact retro-authorizes admissibility;*
5. *coordinates, charts, and frames do not carry physical standing;*
6. *no extra connection, scalar, vector, tensor, boundary, or source structure is silently introduced;*
7. *the projection preserves source-continuation compatibility.*

Definition 14.2 (Metric anchor, tensor, and skin ledger). *In the faithful metric projection:*

- *metric anchors include dimension, signature, differentiability class, boundary regime, metric-compatibility condition, locality envelope, and projection scope;*
- *metric tensors include curvature, causal/geodesic response, transport obstruction, source-response relation, and constraint-compatible dynamics;*
- *metric skin includes coordinates, charts, frames, local labels, gauge representatives, and presentation choices.*

Remark 14.3 (Projection-anchor scope lock). *Four-dimensionality, Lorentzian signature, differentiability class, locality, and minimal metric compatibility are projection anchors in this theorem. They are not generated at the AMetric boundary here. The theorem derives form standing inside this declared projection envelope. A theory that changes these anchors may be legitimate, but it is not an undeclared same-scope alternative to the present minimal first-response result.*

Theorem 14.4 (No primitive metric backflow). *No metric description, curvature equation, or field equation can retro-authorize the standing of the physical interior whose projection it inhabits.*

Proof. A field equation is formulated after the metric projection is admitted. If the

equation is used to fix the admissibility of the projection itself, then a downstream tensor expression is made into a boundary selector. That would either import an anchor not declared in the AASC architecture, privilege a representative, or repair standing post hoc. Each route violates the no-backflow discipline. Therefore metric equations may express standing-bearing projected dynamics, but they cannot authorize the boundary standing of the projection. $\square$

Lemma 14.5 (Diffeomorphism as lawful redescription). *Diffeomorphism covariance is the metric-facing expression of lawful redescription. A coordinate transformation cannot create or destroy standing-bearing physical content.*

Proof. Coordinates and charts are presentation skin unless they change the role content of the theorem object. If a coordinate transformation changes physical content, it is not merely a coordinate transformation; it has moved an anchor or tensor. Lawful redescription preserves quotient identity. Therefore diffeomorphism covariance is the metric projection of quotient identity rather than a primitive metaphysical principle. $\square$

Lemma 14.6 (Minimal metric compatibility). *In the minimal metric projection, the compatible connection satisfies*

$$\nabla_\lambda g_{\mu\nu} = 0.$$

Torsion and metric-affine nonmetricity are not denied; they are richer-scope when introduced.

Proof. The metric role is quotient-stable comparison. The minimal compatible transport preserves that comparison under continuation. The covariant expression of comparison preservation is metric compatibility. If nonmetricity is admitted, the comparison role changes under transport and hence the metric-compatibility anchor has been altered. If torsion is admitted, the connection carries additional tensor content. Both routes may be legitimate, but they are not the minimal same-scope metric projection. $\square$

15 Nonmetric compatibility is not communication

The entanglement arc supplies a general AASC warning: pre-metric compatibility is not interior transport. An entangled preparation is a joint standing class carrying tensor-level compatibility constraints fixed at admission. Later metric descriptions may display spacelike-separated correlations, but no boundary-level transmission occurs.

Definition 15.1 (Boundary compatibility). *Boundary compatibility is tensor-level compatibility fixed at admission before metric distance or time is admissibility-relevant. It is not an interior transport process.*

Theorem 15.2 (No hidden nonmetric transport). *A pre-metric compatibility relation cannot be used as a same-scope transport, repair, or communication channel inside an admitted metric projection. If it is invoked to move standing-bearing content, it either becomes declared tensor load, a boundary-trace certificate, an explicit source/continuation ledger, or inadmissible hidden repair.*

Proof. Transport is an interior continuation relation. Boundary compatibility is fixed prior to metric transport. Treating compatibility as transmission confuses boundary status with interior dynamics. If compatibility is used to alter or move standing content inside the admitted metric projection, it has become tensor load or a continuation ledger and must be declared. If it is not declared, it is hidden repair. Therefore nonmetric compatibility is not nonmetric communication. $\square$

This closes a common escape route. A high-curvature, horizon, or nonlocal model cannot appeal to “nonmetric communication” as a hidden same-scope repair channel. Nonlocal terms are post-admission continuation-envelope expansions, not AMetric compatibility.

16 Metric field equation target class

Definition 16.1 (Metric field-equation candidate). *A metric field-equation candidate is an equation of the form*

$$E_{\mu\nu}[g] = \kappa S_{\mu\nu},$$

where $E_{\mu\nu}[g]$ is a geometric response tensor of the metric projection and $S_{\mu\nu}$ is a claimed source-side tensor ledger.

Definition 16.2 (Same-scope metric candidate). *A metric field-equation candidate is same-scope iff it preserves the anchors, tensor witnesses, quotient identities, continuation rules, and boundary roles of the faithful metric projection. If it changes dimension, signature, connection type, derivative order, source closure, boundary data, nonlocality, or field content, it is richer-scope or effective-scope rather than same-scope.*

Remark 16.3 (Why same-scope does not mean all theories). *Same-scope is a theorem target, not a universal taxonomy. The paper does not assert that higher-curvature, scalar-tensor, torsion, metric-affine nonmetricity, or nonlocal theories are incoherent. It asserts that they are not undeclared minimal first-response metric projections of the already-fixed AASC gravity role.*

Part V

Trajectory Coherence and Boundary-Trace Dynamics

17 AASC coherent trajectory dynamics

Definition 17.1 (Fixed-scope primitive substrate). *A fixed-scope primitive substrate is a tuple*

$$P_D = (X, A, S, \rightsquigarrow, \sim, \text{Ten}),$$

where X is the state-set, A is the admitted scope, S is the standing predicate, $\rightsquigarrow$ is a licensed same-scope transition relation, $\sim$ is standing equivalence, and Ten is tensor content.

Definition 17.2 (Licensed same-scope transition). *A relation $\rightsquigarrow \subseteq X \times X$ is a licensed same-scope transition relation iff $x \rightsquigarrow y$ means that y is reached from x by a same-scope construction move admitted by the fixed role and license structure.*

Remark 17.3 (License is not standing preservation). *A licensed step is a step type within the construction envelope. It does not guarantee that the terminal state preserves standing. This distinction is essential: legality and coherence are different.*

Definition 17.4 (Trajectory and legal trajectory). *A trajectory is a finite nonempty sequence*

$$\tau = (x_0, x_1, \dots, x_n).$$

It is legal iff $x_i \rightsquigarrow x_{i+1}$ for every $i < n$.

Definition 17.5 (Coherent trajectory). *A legal trajectory $\tau = (x_0, \dots, x_n)$ is coherent iff every state on the trajectory has standing:*

$$\text{Coherent}(\tau) \iff \text{Legal}(\tau) \text{ and } \forall i \leq n, S(x_i).$$

Definition 17.6 (Failure witness). *A failure witness for τ is an index $i \leq n$ such that $\neg S(x_i)$.*

Proposition 17.7 (Coherence characterization). *For a legal trajectory τ ,*

$$\neg \text{Coherent}(\tau) \iff \exists i \leq |\tau| \neg S(\tau_i).$$

Proof. This is immediate from the definition of coherence. A coherent legal trajectory has standing at every state; if a state lacks standing, the trajectory is incoherent. Conversely, if no state lacks standing, the legal trajectory is coherent. $\square$

Theorem 17.8 (Composition closure). *If $\text{Coherent}(\tau_1)$, $\text{Coherent}(\tau_2)$, and τ_1, τ_2 are composable, then $\text{Coherent}(\tau_1 \circ \tau_2)$.*

Proof. Legal composition follows from the composability of legal transitions. Standing holds at every state of τ_1 and every state of τ_2 . Since the joining endpoint is shared, every state in the concatenation has standing. Hence the composite trajectory is coherent. $\square$

Theorem 17.9 (Identity-preserving transformation invariance). *Let τ and τ' be legal trajectories that are pointwise identity-preserving equivalent, meaning their corresponding states have equal tensor content. Then*

$$\text{Coherent}(\tau) \iff \text{Coherent}(\tau').$$

Proof. Standing is determined by tensor content on the fixed substrate. If corresponding states have the same tensor content, then standing agrees at each stage. Therefore the universal standing condition defining coherence holds for one trajectory exactly when it holds for the other. $\square$

Corollary 17.10 (Tensor-determined coherence). *Skin-only changes cannot turn an incoherent trajectory into a coherent trajectory. Coherence is determined by tensor content, not presentation.*

Theorem 17.11 (No deferred coherence). *Let τ be a legal trajectory and let τ' be a prefix of τ . If $\neg\text{Coherent}(\tau')$, then $\neg\text{Coherent}(\tau)$.*

Proof. A failure witness in the prefix remains a state of every extension. Since coherence requires standing at every state of the full trajectory, the extension cannot be coherent. Later continuation cannot delete the earlier failure witness. $\square$

Theorem 17.12 (Irrecoverability of coherence failure). *If σ extends τ and $\neg\text{Coherent}(\tau)$, then $\neg\text{Coherent}(\sigma)$.*

Proof. This is the extension form of no deferred coherence. A failure witness in τ is preserved as a witness in σ . Therefore σ cannot be coherent as the same lineage. $\square$

Corollary 17.13 (No coherence by skin repair). *If two legal trajectories are pointwise identity-preserving equivalent and one is incoherent, no skin-level redescription of the other can restore coherence.*

Proof. Identity-preserving equivalence preserves tensor content and hence standing at each stage. An incoherent trajectory contains a standing failure in tensor content. Skin redescription changes no tensor content. Therefore coherence cannot be repaired by skin. $\square$

Theorem 17.14 (Terminal-state criterion). *For legal trajectories in a fixed substrate whose standing is fail-closed along the lineage,*

$$\text{Coherent}(x_0, \dots, x_n) \iff S(x_n).$$

Proof. If the trajectory is coherent, every state has standing, in particular the terminal state. Conversely, if a legal trajectory had an earlier failure witness, downstream propagation of standing failure would prevent the terminal state from standing as the same lineage. Thus terminal standing entails no prior failure witness and hence coherence. $\square$

18 Boundary-trace role fixation

Definition 18.1 (Fixed continuation domain). *A fixed continuation domain is a tuple*

$$D = (E_D, G_D, [L_D]_{\equiv \text{Loc}}, R_D, \text{Aut}_D, \text{Adm}_D),$$

where E_D is the evaluated fragment, G_D is the governance structure, $[L_D]_{\equiv \text{Loc}}$ is a continuation-locus equivalence class, R_D is a role-anchor family, Aut_D is the admissible redescription group, and Adm_D is the admissibility predicate.

Definition 18.2 (Continuation locus). *A continuation locus L_D is the minimally preserved admissibility-relevant persistence structure induced by same-domain standing-preserving continuation.*

Definition 18.3 (Role-anchor slot). *A role-anchor slot $\text{Slot}_D(L)$ is the admissibility-bearing position by which a continuation locus L participates in a fixed constructional role.*

Definition 18.4 (Role-anchor equivalence). *Two role anchors R, R' are role-equivalent, written $R \equiv_{\text{Role}} R'$, iff they preserve identical admissibility-bearing role behavior under lawful same-domain continuation, redescription, boundary comparison, and failure handling.*

Definition 18.5 (Internal non-selection). *$\text{InternalNonSel}_D(S_L)$ holds iff no admissibility-bearing structure available inside the fixed-domain role problem uniquely privileges one candidate role anchor over all others without adding selector load or changing scope.*

Definition 18.6 (Boundary trace). *A boundary trace B for a candidate role-anchor family S_L is an admissibility-preserving asymmetry interface between internally non-selective continuation structure and role-anchor fixation. It is bivalent, stable under identity-preserving redescription, and declared as part of the fixed target domain.*

Definition 18.7 (Boundary-trace fixation).

$$\text{Fix}_D(B, [L_D]_{\equiv \text{Loc}}, [R_L]_{\equiv \text{Role}})$$

holds iff the admissible boundary trace B fixes the role-anchor equivalence class $[R_L]_{\equiv \text{Role}}$ for the continuation-locus class $[L_D]_{\equiv \text{Loc}}$, without selecting a representative R_L .

Theorem 18.8 (AMetric index non-selection). *At an AMetric boundary, no canonical index, token address, provenance marker, channel label, representative name, probability, metric, or ranking fixes a role-anchor class by default.*

Proof. The AMetric boundary contains no admitted selector, metric, ranking, or representative privilege. Any attempted default either remains inert or introduces admitted structure. If admitted structure is introduced, the boundary is no longer doing the selection; the declared structure is. Therefore no AMetric default selects a role anchor. $\square$

Theorem 18.9 (Boundary-trace fixation is not representative selection). *Boundary-trace fixation of a role-anchor class neither selects a unique representative nor collapses to a selector operation:*

$$\exists! [R_L]_{\equiv \text{Role}} \not\Rightarrow \exists! R_L.$$

Proof. The boundary trace is defined on role behavior up to $\equiv_{\text{Role}}$. If the positive boundary orbit is a singleton equivalence class, the role is fixed. Representatives inside that class remain equivalent under lawful redescription. Selecting one would require a further selector not supplied by the trace. Therefore boundary-trace fixation fixes a class, not a representative. $\square$

Theorem 18.10 (Boundary-trace normal form). *Every admissibility-preserving fixation regime for an internally non-selective continuation-locus role problem factorizes through a declared boundary trace, declared internal asymmetry, declared provenance/indexical asymmetry, or a scope-changing enrichment. No fourth same-domain authority exists.*

Proof. If internal structure fixes the role, it is declared internal asymmetry. If provenance or indexical structure fixes the role, it is declared provenance/indexical asymmetry. If neither is available but a stable bivalent boundary predicate fixes the role class, it is boundary-trace fixation. If none of these fixes the role and a unique role is nevertheless

claimed, the claim imports an undeclared selector or changes scope. Hence the listed routes exhaust same-domain fixation. $\square$

Proposition 18.11 (Metric/source/vacuum/horizon role fixation). *In the present metric-projection theorem, the metric role, source-ledger role, homogeneous vacuum-ledger role, and horizon-certificate role must be fixed as role-anchor equivalence classes, not as primitive representatives.*

Proof. Each role participates in standing-bearing continuation. The metric role fixes comparison and transport; the source role fixes standing-bearing source response; the vacuum role fixes homogeneous baseline content; the horizon-certificate role fixes threatened standing preservation under high-curvature boundary conditions. If any were selected by representative privilege, the theorem would import a hidden selector. Boundary-trace fixation provides role-anchor class fixation without representative selection. $\square$

Role	Why representative selection fails	Boundary-trace fixation
Metric role	Coordinates, frames, foliations, and representative metric descriptions cannot carry primitive standing.	Fixes the metric role-anchor class for comparison and compatible transport.
Source role	Scalar mass, energy labels, or matter bookkeeping do not by themselves fix metric-facing source standing.	Fixes the closed source-ledger class whose exchanges are included or declared open/effective.
Vacuum role	The split (Λ_R, ρ_R) is not quotient-stable under homogeneous shifts.	Fixes the homogeneous effective vacuum ledger represented by $\Lambda g_{\mu\nu}$.
Horizon/certificate role	Interior retention, global storage, or remnant assertion does not preserve exterior standing by representative choice.	Fixes certificate, leakage/radiation, endpoint-leakage, or non-sink quotient roles when high-curvature standing is threatened.

19 Closed metric/source trajectory theorem

Definition 19.1 (Metric/source trajectory). *A metric/source trajectory is a legal trajectory in the faithful metric projection whose states include metric tensor content, source-ledger*

content, quotient identity, boundary-role status, and continuation relation. Write it as

$$\tau_{g,T} = ((g_0, T_0), \dots, (g_n, T_n)).$$

Definition 19.2 (Closed same-scope source ledger). *A source ledger $T_{\mu\nu}^{\text{closed}}$ is closed at the declared scope iff every standing-bearing source exchange relevant to metric response is included in the tensor ledger or declared as a boundary/effective exchange. No source flux is hidden outside the ledger. When this closed ledger is represented through metric-response extraction, its local variation is written*

$$\delta_g \mathcal{L}_{\text{source}}^{\text{closed}} = -\frac{1}{2} \sqrt{-g} T_{\mu\nu}^{\text{closed}} \delta g^{\mu\nu}.$$

This extraction formula records the metric-facing tensor type of the already admitted source ledger; it is not an action-primitivity assumption.

Definition 19.3 (Internal source equations). *Internal source equations are the standing-preserving continuation equations of the source ledger's own roles inside the declared closed source scope. They include, when present, particle-persistence equations, field-carrier mediation equations, deformation-response and inertia transport equations, interaction/restoration equations, and vacuum-baseline continuation equations. They are not primitive matter-action assumptions; they are the source-side continuation conditions that keep the closed ledger from gaining or losing standing-bearing source content without a declared exchange, anomaly, boundary flux, open/effective scope, or ledger incompleteness.*

Theorem 19.4 (Closed source ledger coherence). *If a metric/source trajectory is coherent and the source ledger is closed at the declared same-scope metric projection, then*

$$\nabla^\mu T_{\mu\nu}^{\text{closed}} = 0.$$

Proof. A coherent trajectory preserves standing at every stage. The closed source ledger contains all source content whose exchange affects metric response. If the covariant divergence were nonzero without a declared exchange ledger, then source content would leave or enter the metric-facing ledger without a standing-bearing transition. That would be either hidden tensor load, an undeclared boundary channel, or standing failure. Any of those supplies a failure witness for the source trajectory. Since the trajectory is coherent, no such witness is present. Therefore the closed same-scope source ledger is divergence-compatible. $\square$

Corollary 19.5 (Open and effective source caveat). *If $\nabla^\mu T_{\mu\nu}^{\text{closed}} \neq 0$ in a presented model, then at least one of the following holds: the source ledger is incomplete, an exchange ledger is omitted, the system is open or effective, anomaly/failure is present, or the scope has changed. It is not a counterexample to the closed same-scope theorem.*

Remark 19.6 (Relation to the ordinary GR conservation statement). *In conventional GR language, the closed-ledger condition corresponds to the familiar covariant conservation of stress-energy. The present argument does not import that condition from GR. It retypes it as the metric projection of coherent closed source continuation. A geometric response with nonzero covariant divergence cannot be matched to a closed source ledger without hidden exchange, source incompleteness, anomaly, or scope change.*

Proposition 19.7 (Diffeomorphism-facing source-closure bridge). *In a faithful metric projection, suppose a closed source ledger admits a metric-facing extraction functional that is invariant under lawful redescription/diffeomorphism, and suppose the standing-bearing source roles satisfy their internal source equations within the declared closed ledger. Then the extracted tensor is covariantly divergence-compatible,*

$$\nabla^\mu T_{\mu\nu}^{\text{closed}} = 0,$$

modulo a declared exchange ledger, anomaly, boundary flux, open/effective scope, or ledger incompleteness.

Proof. Lawful redescription is represented metrically by diffeomorphism covariance. Varying the closed source extraction by an infinitesimal redescription gives a term proportional to the source equations plus a term whose local coefficient is $\nabla^\mu T_{\mu\nu}^{\text{closed}}$, up to boundary or exchange contributions. By the preceding definition, the source-equation term vanishes exactly when the ledger roles preserve standing under their own declared continuation equations. If no boundary flux, anomaly, open/effective exchange, or omitted ledger is declared, the coefficient of the arbitrary redescription vector must vanish, giving $\nabla^\mu T_{\mu\nu}^{\text{closed}} = 0$. If it does not vanish, the nonzero term is precisely a declared exchange, anomaly, boundary flux, open/effective scope, or ledger incompleteness. This is a conventional-facing compatibility witness for AASC source closure, not an import of the Einstein equation. $\square$

Theorem 19.8 (Geometric response compatibility). *For a field equation*

$$E_{\mu\nu} = \kappa T_{\mu\nu}^{\text{closed}},$$

closed source coherence requires

$$\nabla^\mu E_{\mu\nu} = 0.$$

Proof. Taking the covariant divergence of both sides gives $\nabla^\mu E_{\mu\nu} = \kappa \nabla^\mu T_{\mu\nu}^{\text{closed}}$. The right-hand side vanishes by closed source ledger coherence. Therefore the geometric response is divergence-compatible. If it were not, the field equation would create or destroy source standing by geometric mismatch. $\square$

Part VI

Same-Scope First-Response Metric Candidates

20 The predicate $\text{SameScopeMetric}_1(E_{\mu\nu})$

Theorem 20.1 (SameScopeMetric₁ as metric target translation). *The predicate SameScopeMetric₁($E_{\mu\nu}$) is the metric-response translation of the fixed AASC target, not a list tailored after the conclusion. Each clause blocks a specific failure of the declared theorem object.*

Proof. Metric internality blocks anchor motion outside the licensed projection. Tensoriality and the ban on representative privilege block skin from carrying force. Rank, symmetry, and closed-source compatibility match the metric-facing source ledger type. Locality blocks undeclared continuation-envelope import. First-response order blocks higher-response tensor load. The exclusions of curvature products, curvature derivatives, arbitrary functions, extra fields, torsion, metric-affine nonmetricity, nonlocal operators, and boundary/global data block undeclared richer-scope structure. The no-source-laundrying clause blocks primitive source import on the geometric side. Thus the predicate is the formal metric image of fixed anchors, declared tensor content, no skin authority, and closed-source compatibility. $\square$

Predicate burden	Failure blocked
Metric internality	Anchor motion outside (M, g, ∇) .
Tensoriality and no representative privilege	Coordinate, frame, or skin-level force.
Rank and symmetry	Type mismatch with the closed metric-facing source ledger.
Locality	Hidden nonlocal continuation import.
First-response order	Higher-response/effective tensor load.
No extra fields, torsion, nonmetricity	Undeclared field or connection-scope enrichment.
No source-side laundrying	Primitive source content placed on the geometric side.
Closed-source compatibility	Geometric/source continuation mismatch.
Nontrivial general source response	Pure homogeneous baseline masquerading as full matter response.

Definition 20.2 (First-response metric content). *First-response metric content is the lowest-order local tensorial obstruction to compatible metric continuation inside the licensed metric projection. For the minimal Levi-Civita projection, the first such obstruction is curvature. After the rank-two source-response target is fixed, first-response metric content can enter only through the Ricci response, scalar-trace return, and metric baseline slots identified below.*

Definition 20.3 (Minimal same-scope first-response metric predicate). *Write $\text{SameScopeMetric}_1(E_{\mu\nu})$ for $\text{SameScopeMetric}_1(E_{\mu\nu})$. The predicate holds iff $E_{\mu\nu}$ satisfies the following clauses:*

- (S1) **Metric internality.** $E_{\mu\nu}$ is built only from the licensed metric projection (M, g, ∇) , where ∇ is the minimal metric-compatible connection.
- (S2) **Tensoriality.** $E_{\mu\nu}$ is tensorial under lawful redescription/diffeomorphism.
- (S3) **Rank type.** $E_{\mu\nu}$ is a rank-two covariant tensor.
- (S4) **Symmetry.** $E_{\mu\nu} = E_{\nu\mu}$.
- (S5) **Locality.** $E_{\mu\nu}$ is local in the metric projection.
- (S6) **First-response order.** $E_{\mu\nu}$ is linear in the first admitted curvature obstruction.
- (S7) **No curvature products.** It contains no R^2 , $R_{\mu\nu}R^{\mu\nu}$, Weyl-squared, or other higher-response product.
- (S8) **No curvature derivatives.** It contains no ∇R , $\square R$, derivative of curvature, or higher continuation memory.
- (S9) **No arbitrary function selectors.** It contains no $f(R)$, arbitrary curvature function, or parameterized selector unless declared richer-scope.
- (S10) **No extra fields.** It contains no scalar, vector, spinor, tensor, matter, or auxiliary field on the geometric side.
- (S11) **No torsion.** Torsion is absent unless declared as richer-scope connection load.
- (S12) **No metric-affine nonmetricity.** Metric-affine nonmetricity is absent unless declared as an altered metric-compatibility anchor.
- (S13) **No nonlocal operators.** It contains no inverse wave operator, integral kernel, global constraint, or nonlocal continuation import.
- (S14) **No boundary/global data.** It contains no boundary term, asymptotic selector, topology selector, or global datum as local same-scope response.
- (S15) **No source-side laundering.** Source content cannot be hidden on the geometric side.
- (S16) **No representative privilege.** Coordinate/frame-dependent expressions are skin.
- (S17) **Closed-source compatibility.** It is capable of matching a closed same-scope source ledger.

(S18) **Nontrivial general source response.** *For a general closed source ledger, the candidate is not merely a zero-curvature homogeneous baseline; its first-curvature response branch is active. The pure term $cg_{\mu\nu}$ is allowed as homogeneous vacuum/cosmological ledger content, not as the full response equation for arbitrary matter.*

Remark 20.4 (Target of the predicate). *The predicate SameScopeMetric₁ does not classify all possible gravitational theories. It classifies the minimal same-scope first-response metric class. In ATS terms it requires fixed anchors, declared tensor content, and no skin carrying standing-bearing force. A theory outside SameScopeMetric₁ may be legitimate as richer-scope, effective-scope, horizon-certified, semiclassical, or nonlocal, but it is not the theorem target. This is not definition by exclusion: the predicate is the formal expression of the already-fixed AASC target class—no anchor motion, no undeclared tensor load, no skin authority, first admitted curvature response, closed source compatibility, and no boundary repair. Higher-curvature or modified terms are excluded from SameScopeMetric₁ because they add higher-response or richer-scope structure, not because they are mathematically invalid.*

21 Six-lemma curvature-response exhaustion

Before the exhaustion proof begins, fix the comparator boundary. This argument is not Lovelock repackaged. Lovelock-style uniqueness classifications operate after the metric framework and variation assumptions are already admitted. The AASC proof first licenses the metric projection from AMetric non-selection, fixes source and continuation standing, and defines the same-scope first-response predicate SameScopeMetric₁. External uniqueness theorems are useful comparators; they are not proof authority here.

Lemma 21.1 (Tensoriality). *Any connection-only expression fails same-scope metric standing unless completed into curvature.*

Proof. The connection coefficients of the metric-compatible connection are not tensorial under arbitrary redescription. A connection-only expression therefore depends on presentation unless it is assembled into a tensorial obstruction. The first such obstruction is curvature. If a connection-only expression is treated as standing-bearing before being completed into a tensor, it privileges a representative and violates quotient identity. Therefore connection-only terms fail SameScopeMetric₁. $\square$

Lemma 21.2 (Rank type). *A metric field equation coupled to a closed symmetric source ledger requires a symmetric rank-two metric-facing tensor on the geometric side.*

Proof. The closed source ledger is metric-facing and has type $(0, 2)$. A direct equation requires matching type. A scalar cannot couple without being returned to rank two; a higher-rank curvature tensor cannot couple without contraction or additional tensor structure; an antisymmetric part would require torsion, spin-current coupling, or a separate connection ledger. These are richer-scope unless declared. Hence the minimal same-scope geometric response is symmetric rank two. $\square$

Lemma 21.3 (First-curvature slot). *The first rank-two contraction of the curvature obstruction of the minimal metric connection is the Ricci tensor $R_{\mu\nu}$.*

Proof. The curvature of the minimal metric connection has Riemann type $R^\rho{}_{\sigma\mu\nu}$. A rank-two tensor must contract two indices using metric-internal structure. The standard contraction yields $R_{\mu\nu} = R^\rho{}_{\mu\rho\nu}$. Other metric-internal contractions either reproduce Ricci/scalar curvature under the symmetries of curvature, vanish, or require extra tensor structure to return the correct type. The trace-free Ricci tensor is not an independent first-response slot; in dimension n ,

$$R_{\mu\nu}^{TF} = R_{\mu\nu} - \frac{1}{n}Rg_{\mu\nu},$$

and in four dimensions this is $R_{\mu\nu}^{TF} = R_{\mu\nu} - \frac{1}{4}Rg_{\mu\nu}$, already inside the span of the Ricci and scalar-trace slots. Extra structure violates SameScopeMetric₁ unless declared richer-scope. Therefore $R_{\mu\nu}$ is the first rank-two curvature slot. $\square$

Lemma 21.4 (Trace return). *The scalar trace of the first curvature response is $R = g^{\mu\nu}R_{\mu\nu}$, and the only same-scope metric-internal way to return that scalar to rank-two form is $Rg_{\mu\nu}$.*

Proof. The scalar trace R is metric-internal and redescription invariant but has rank zero. The equation target is rank two. The only metric-internal rank-two baseline available without adding a new tensor is $g_{\mu\nu}$. Therefore the scalar trace can enter a same-scope rank-two geometric response only as $Rg_{\mu\nu}$. Any alternative requires an additional tensor, preferred direction, source field, boundary datum, or representative selector. $\square$

Lemma 21.5 (Metric baseline). *The only zero-curvature same-scope rank-two metric baseline is $g_{\mu\nu}$, up to constant coefficient.*

Proof. A zero-curvature geometric term contains no curvature response. It must still be metric-internal, rank two, symmetric, and redescription invariant. The metric itself supplies exactly that baseline. A scalar constant multiple $cg_{\mu\nu}$ is allowed as homogeneous baseline content. A nonconstant scalar multiplier requires an extra field, boundary datum, or selector and is therefore outside SameScopeMetric₁. $\square$

Lemma 21.6 (Orientation, parity, and topological terms). *A term requiring an orientation tensor, density, parity-odd contraction, Pontryagin/Euler density, or topological boundary datum is not a minimal same-scope first-response symmetric rank-two metric tensor unless the relevant orientation, density, topological, or boundary anchor is declared. If declared, it is boundary/global or richer-scope structure rather than an element of SameScopeMetric₁.*

Proof. The predicate SameScopeMetric₁ permits only metric-internal, local, first-response, symmetric rank-two tensor content. Orientation tensors and density terms introduce an additional orientation or volume-form anchor beyond the bare metric response. Topological densities and boundary terms control global or boundary ledgers rather than the local first-response metric side. If such structure has standing-bearing force, it must be declared as an anchor or tensor load; if it has no standing-bearing force, it is skin. It therefore does not supply an undeclared same-scope alternative to the Ricci, scalar-trace, and metric-baseline slots. $\square$

Lemma 21.7 (Exclusion). *Every candidate outside $R_{\mu\nu}$, $Rg_{\mu\nu}$, and $g_{\mu\nu}$ either fails SameScopeMetric₁ or is declared richer-scope/effective-scope.*

Proof. Curvature products are higher-response terms. Curvature derivatives import higher continuation memory. Arbitrary functions of curvature import selectors and typically extra scalar response. Torsion changes the connection tensor ledger. Metric-affine nonmetricity changes the metric-compatibility anchor. Extra fields add tensor/source load. Nonlocal terms expand the continuation envelope. Boundary and global terms are anchors, not local same-scope response. Coordinate terms are skin. Quantum expectation terms require mixed-projection record closure. Horizon endpoint terms require certificate, lawful leakage or radiation channel, independently licensed endpoint-leakage certificate, non-sink quotient, or richer scope. No candidate outside the three slots remains a minimal same-scope first-response metric tensor. $\square$

Remark 21.8 (Scope of first-response exclusion). *The exclusion of curvature products, curvature derivatives, nonlocal operators, torsion, metric-affine nonmetricity, and extra fields is a scope classification, not a denial that such terms can be written or can be physically meaningful in declared richer/effective regimes. They do not belong to the minimal first-response same-scope metric class unless they collapse to the linear first-response form.*

Theorem 21.9 (Curvature-response normal form). *If SameScopeMetric₁($E_{\mu\nu}$), then*

$$E_{\mu\nu} = aR_{\mu\nu} + bRg_{\mu\nu} + cg_{\mu\nu}$$

where a, b, c are projection-envelope coefficients fixed within the declared metric scope. They are not AMetric defaults or boundary-generated numerical values; their dimensional interpretation requires source coupling, scale anchoring, or calibration.

Proof. By tensoriality, the first response must be curvature rather than connection-only presentation. By the rank-type lemma, it must be symmetric rank two. By the first-curvature slot lemma, the rank-two curvature slot is $R_{\mu\nu}$. By the trace-return lemma, the scalar trace slot is $Rg_{\mu\nu}$. By the metric-baseline lemma, the zero-curvature baseline slot is $g_{\mu\nu}$. By the exclusion lemma, all other candidates fail SameScopeMetric₁ or leave the theorem scope. Therefore the most general same-scope minimal first-response candidate is the stated linear combination. $\square$

Table 4: Main-text candidate tensor challenge ledger.

Candidate	Disposition inside the minimal first-response target
Trace-free Ricci or Schouten tensor	Linear recombinations of $R_{\mu\nu}$ and $Rg_{\mu\nu}$; not independent first-response slots.
Weyl tensor	First-curvature content, but no direct symmetric rank-two source-response tensor without contraction, differentiation, squaring, projection, or auxiliary structure.
Bach tensor	Curvature-derivative/higher-response object; outside first-response order.
Cotton-type terms	Dimension-specific derivative or orientation-sensitive structures; not same-scope four-dimensional first-response metric tensors.
Gauss-Bonnet/Euler density	Boundary/topological or higher-scope structure in the declared four-dimensional local first-response class; no undeclared local rank-two alternative.
Pontryagin or parity-odd terms	Require orientation, density, topological, or boundary anchors; not metric-internal first-response tensors unless declared richer scope.
Nonconstant scalar multipliers	Scalar selector, boundary datum, or extra field unless declared; constant multiples are already the coefficients a, b, c .
$f(R)$ and arbitrary functions	Function selectors and usually scalar-response dynamics; collapse to linear first-response only when reduced to the normal form.
Curvature products/derivatives	Higher-response or higher-continuation-memory tensor load.

Candidate	Disposition inside the minimal first-response target
Extra scalar, vector, tensor, torsion, or metric-affine terms	Declared richer-scope structure, not undeclared same-scope first response.

22 Relation to metric uniqueness classifications

This argument is not an invocation of Lovelock’s theorem or any other external metric uniqueness theorem. Metric-uniqueness classifications operate after a metric framework and its differentiability assumptions have already been admitted. The AASC problem is prior: it asks how metric response obtains standing from the AMetric boundary, how metric projection is role-fixed without primitive representative selection, which source roles are licensed as metric-facing tensor content, and why same-scope first-response closure admits only the Ricci response, scalar-trace response, and metric baseline before source-continuation compatibility fixes the Einstein combination. Once the AASC derivation reaches a faithful four-dimensional Lorentzian metric projection, its result is consistent with familiar metric uniqueness classifications. Those classifications are not used as premises. The AASC proof burden is admissibility, standing, boundary-trace role fixation, source-ledger licensing, and same-scope first-response exhaustion.

Part VII

Source Continuation and Einstein Combination

23 Divergence compatibility

Theorem 23.1 (Einstein-combination theorem). *Let*

$$E_{\mu\nu} = aR_{\mu\nu} + bRg_{\mu\nu} + cg_{\mu\nu}.$$

If $\nabla^\mu E_{\mu\nu} = 0$ for general curvature response in the faithful metric projection, then $b = -a/2$. In the nontrivial general source-response branch $a \neq 0$,

$$E_{\mu\nu} = aG_{\mu\nu} + cg_{\mu\nu}.$$

Proof. The licensed metric projection supplies the internal curvature identity

$$\nabla^\mu R_{\mu\nu} = \frac{1}{2}\nabla_\nu R$$

and metric compatibility

$$\nabla^\mu g_{\mu\nu} = 0.$$

The contracted curvature identity is not an Einstein-equation premise; it is an internal identity of the admitted Levi-Civita metric grammar. It fixes coefficient discipline only after the metric projection has been licensed. Therefore

$$\begin{aligned}\nabla^\mu E_{\mu\nu} &= a\nabla^\mu R_{\mu\nu} + b\nabla^\mu(Rg_{\mu\nu}) + c\nabla^\mu g_{\mu\nu} \\ &= a\left(\frac{1}{2}\nabla_\nu R\right) + b\nabla_\nu R + 0 \\ &= \left(\frac{a}{2} + b\right)\nabla_\nu R.\end{aligned}$$

For general curvature response this vanishes iff $a/2 + b = 0$. Hence $b = -a/2$. Thus

$$E_{\mu\nu} = a\left(R_{\mu\nu} - \frac{1}{2}Rg_{\mu\nu}\right) + cg_{\mu\nu} = aG_{\mu\nu} + cg_{\mu\nu}.$$

□

Proposition 23.2 (Degenerate homogeneous branch). *If the normal form has $a = 0$ and is required to be divergence-compatible for general curvature response, then $b = 0$ and $E_{\mu\nu} = cg_{\mu\nu}$. This branch is a homogeneous baseline ledger. It may contribute the vacuum/cosmological term, but it is not the full metric response equation for a general closed matter source. The Einstein-source theorem therefore uses the nontrivial first-curvature response branch $a \neq 0$.*

Proof. With $a = 0$, the divergence calculation gives $\nabla^\mu E_{\mu\nu} = b\nabla_\nu R$. General curvature response forces $b = 0$. The surviving term $cg_{\mu\nu}$ is zero-curvature homogeneous baseline content. It can be absorbed into the cosmological ledger after normalization in the nontrivial branch, but it cannot match arbitrary closed source deformation-response. □

Corollary 23.3 (Closed-source Einstein form). *For a closed same-scope source ledger, a field equation with $\text{SameScopeMetric}_1(E_{\mu\nu})$ in the nontrivial first-curvature response branch*

has the form

$$G_{\mu\nu} + \Lambda g_{\mu\nu} = \kappa T_{\mu\nu}^{\text{closed}}.$$

Proof. The closed-source theorem gives $\nabla^\mu T_{\mu\nu}^{\text{closed}} = 0$, hence $\nabla^\mu E_{\mu\nu} = 0$. The normal-form theorem and Einstein-combination theorem give $E_{\mu\nu} = aG_{\mu\nu} + cg_{\mu\nu}$. Coupling to the source ledger and normalizing constants yields the stated equation with $\Lambda = c/a$ and κ the source-geometry conversion factor. $\square$

24 Stress-energy as source ledger

The stress-energy tensor is not assumed as primitive mass-energy substance. It is the metric-facing extraction of already admitted source roles under a closed source ledger. Likewise, $\nabla^\mu T_{\mu\nu}^{\text{closed}} = 0$ is not imported backward from GR; it is the metric projection of coherent closed source continuation.

Source-order stage	Status
Standing-bearing source roles	Particle persistence, deformation-response, field-carrier mediation, interaction, transport, restoration, and vacuum-baseline roles are fixed first.
Closed source ledger	All metric-relevant exchanges are included or declared open/effective.
Metric-facing extraction	The tensor $T_{\mu\nu}^{\text{closed}}$ records how the closed ledger responds to metric variation.
Action notation	Optional extraction formalism where an action presentation is available; not a primitive source premise.

Definition 24.1 (Metric-facing source extraction). *A closed source ledger admits a stress-energy tensor when its metric response satisfies*

$$\delta_g \mathcal{L}_{\text{source}}^{\text{closed}} = -\frac{1}{2} \sqrt{-g} T_{\mu\nu}^{\text{closed}} \delta g^{\mu\nu}.$$

If an action formalism is used, this is equivalently written

$$T_{\mu\nu}^{\text{closed}} = -\frac{2}{\sqrt{-g}} \frac{\delta S_{\text{source}}^{\text{closed}}}{\delta g^{\mu\nu}}.$$

The shorter notation $T_{\mu\nu}$ may be used only after closure and scope are fixed. This is an extraction formalism, not a primitive action postulate.

Definition 24.2 (AASC source ledger). *An AASC source ledger is the metric-facing tensor representation of standing-bearing non-geometric roles that affect metric response: particle persistence classes, deformation-response profiles, inertia/transport response, field-carrier mediation, interaction terms, restoration and perturbation burden, vacuum-baseline extraction, and boundary/effective terms when declared.*

Theorem 24.3 (Source ledger admissibility). *A source tensor is admissible in the faithful metric projection iff it is metric-facing, symmetric, redescription invariant, closed or declared open/effective, extracted from standing-bearing source roles, and not primitive scalar mass, primitive energy ledger, hidden vacuum repair, or undeclared boundary import.*

Proof. Metric-facing source response must have the same tensor type as the metric variation. Redescription invariance prevents source content from depending on coordinates or presentation. Closure prevents hidden flux. Standing-bearing extraction prevents a scalar label, energy ledger, or vacuum number from acquiring source status without role content. If any of these conditions fails, the source is either skin, hidden tensor load, open/effective scope, or standing failure. $\square$

25 Mass/inertia bridge

Definition 25.1 (Particle persistence class). *A particle role is a localized persistence class*

$$\Pi_D = [\kappa_U]_{\equiv \Pi}.$$

Definition 25.2 (Mass and inertia role). *Mass is the deformation-response profile*

$$\mu_D(\Pi) : \Delta_\Pi \rightarrow C_D,$$

and inertia is its transport-facing restriction

$$\iota_D(\Pi) = \mu_D(\Pi)|_{\Delta_\Pi^{tr}}.$$

Proposition 25.3 (Mass-to-source bridge). *The metric-facing source contribution of localized matter follows the role chain*

$$\Pi_D \longrightarrow \mu_D(\Pi) \longrightarrow \iota_D(\Pi) \longrightarrow T_{\mu\nu}^{particle/source}.$$

Proof. A particle persistence class supplies the identity target. The mass profile supplies the standing-preserving deformation response of that target. The inertia profile supplies the transport-state response. The metric projection requires a source ledger representing

deformation and transport response as a metric-facing tensor. Thus the particle contribution to $T_{\mu\nu}$ is not a primitive scalar mass; it is the stress-energy extraction of already-licensed deformation-response content. $\square$

Corollary 25.4 (No primitive mass source). *A scalar mass assignment is gravitationally source-bearing only if it represents licensed deformation-response inside the closed source ledger.*

Part VIII

Vacuum, ZPE, and the Cosmological Term

26 Vacuum baseline

Definition 26.1 (Vacuum profile). *A vacuum profile over a local continuation domain is*

$$\Omega_D(U) = (\Phi_U^0, \partial_U^0, T_U^0, Q_U^0, G_U^0, \mu_U^0, R_U^0).$$

It is a non-null no-particle baseline retaining field-carrier, boundary, transport, quotient comparison, gauge, deformation-response, and restoration roles.

Proposition 26.2 (No-particle is not nullity).

$$NoPartExc_D(\Omega_D) \not\approx Null_D.$$

Proof. The no-particle condition removes localized particle-class excitations. It does not remove the field-carrier baseline, boundary trace, transport possibility, quotient comparison, gauge compatibility, restoration, or deformation-response baseline. Therefore it is not a nullity condition. $\square$

Theorem 26.3 (No primitive vacuum-energy source). *A scalar vacuum-energy assignment does not carry source standing until the vacuum role and extraction ledger are fixed.*

Proof. A scalar number alone does not define carrier baseline, boundary trace, transport possibility, quotient comparison, gauge compatibility, restoration, or deformation response. If it enters after a metric/source extraction ledger is fixed, it may represent projected vacuum content. If introduced before that, it is selector import or bookkeeping. $\square$

27 ZPE quotient discipline

Definition 27.1 (Homogeneous vacuum sector). *The homogeneous renormalized vacuum sector has the metric-proportional form*

$$\langle T_{\mu\nu} \rangle_{0,R} = -\rho_R g_{\mu\nu}.$$

Definition 27.2 (Effective cosmological combination). *The metric-facing effective homogeneous term is represented by*

$$\Lambda_{eff} = \Lambda_R + 8\pi G_R \rho_R.$$

The separate split (Λ_R, ρ_R) is not standing-bearing without further extraction structure.

Theorem 27.3 (Homogeneous shift invariance). *The transformation*

$$\rho_R \mapsto \rho_R + \delta\rho, \quad \Lambda_R \mapsto \Lambda_R - 8\pi G_R \delta\rho$$

preserves Λ_{eff} . Therefore the absolute homogeneous zero-point coefficient is not a standalone prediction of the cosmological term.

Proof. Substitution gives

$$\Lambda_R - 8\pi G_R \delta\rho + 8\pi G_R (\rho_R + \delta\rho) = \Lambda_R + 8\pi G_R \rho_R.$$

Thus the effective metric-facing term is unchanged. A quantity that changes under this admissible homogeneous redescription lacks standalone standing. Only the quotient-stable combination is standing-bearing. $\square$

Corollary 27.4 (Cosmological term as quotient-stable homogeneous ledger). *The term $\Lambda g_{\mu\nu}$ is licensed as a quotient-stable homogeneous metric ledger. It is not licensed as primitive zero-point vacuum prediction.*

Remark 27.5 (No numerical claim). *The theorem derives the admissible form of the homogeneous metric term. It does not compute Λ , solve the observed cosmic acceleration problem, or identify dark energy as a primitive substance.*

Vacuum/cosmological claim	Status here
$\Lambda g_{\mu\nu}$ as homogeneous metric ledger	Licensed.
Cutoff zero-point sum predicts Λ	Not licensed.
Absolute numerical Λ generated at AMetric boundary	Not reported.
Cosmological extraction/calibration needed	Required for numerical value.

Part IX

High Curvature and Horizon Closure

28 Purpose of the horizon module

The horizon module is not used to derive the Einstein equation. It is route-closure discipline: high-curvature or horizon structure cannot be smuggled back as hidden same-scope repair. It closes an escape route: one might argue that high curvature allows hidden same-scope repair terms beyond the minimal first-response equation. AASC blocks that move. High-curvature responses must be declared as effective/richer-scope tensor load, lawful leakage or radiation channel, independently licensed endpoint-leakage certificate, boundary certificate, non-sink quotient, or scope expansion. They cannot function as hidden same-scope repair.

Definition 28.1 (Exterior horizon reconstruction scope). *An exterior horizon reconstruction scope is a tuple*

$$S_{hor} = (E_{ext}, BH, R_{ext}, W_{ext}, Cont, Env),$$

where E_{ext} is the exterior evaluated domain, BH is the horizon/trapped-region presentation, R_{ext} is the exterior reconstruction relation, W_{ext} is the exterior witness algebra, $Cont$ is the continuation relation, and Env is the admissibility envelope.

Definition 28.2 (Certificate-complete horizon scope). *A horizon scope is certificate-complete when every preservation of threatened standing is represented through at least one of: exterior witness algebra, leakage channel, standing-transport certificate, or non-sink quotient.*

Theorem 28.3 (No hidden horizon repair). *A high-curvature or horizon response is admissible on the same AASC metric-projection scope only if it is one of:*

1. *skin-equivalent presentation of exterior standing content;*

2. *declared high-curvature tensor/effective load;*
3. *lawful leakage or radiation channel, or an independently licensed endpoint-leakage certificate;*
4. *boundary-to-boundary standing certificate;*
5. *non-sink quotient reconstruction;*
6. *declared richer-scope theory.*

Otherwise it is hidden repair, global retention laundering, remnant compression, selector import, or scope drift.

Proof. A horizon threatens exterior standing when it blocks reconstruction, transfer, or certification of standing-bearing content. If the claimed preservation changes no exterior standing, it is skin. If it changes standing by additional curvature/source terms, those terms are declared tensor load. If standing exits by a lawful leakage or radiation channel, or by an independently licensed endpoint-leakage certificate, that channel is part of the continuation ledger. If standing is reconstructed at the boundary, a certificate or non-sink quotient is present. If none of these holds but standing is claimed, the claim either stores standing in an inaccessible remnant, appeals to global retention without exterior reconstruction, imports a selector, or changes scope. None is same-scope hidden repair. $\square$

Corollary 28.4 (No standing sinks). *A permanent horizon cannot function as a same-scope standing sink. If it threatens exterior standing, it must provide certificate, lawful leakage or radiation channel, independently licensed endpoint-leakage certificate, non-sink quotient, or richer-scope structure.*

Corollary 28.5 (High-curvature corrections). *Higher-curvature terms may be admissible as declared effective/high-curvature tensor load. They are not same-scope first-response replacements for Einstein dynamics unless they collapse to the Einstein form.*

Part X

Modified Gravity Classification

Corollary 28.6 (No horizon rescue by nonmetric communication). *A horizon or high-curvature endpoint cannot preserve exterior standing by appealing to nonmetric communication. A Metric compatibility is non-transmissive. A standing-transfer claim must supply exterior witness algebra, leakage/radiation, boundary certificate, non-sink quotient, or declared richer-scope tensor load.*

29 Classification theorem

Outside SameScopeMetric₁ does not mean false. It means not an undeclared minimal first-response same-scope alternative. Declared richer-scope and effective theories may be legitimate; they are simply not the theorem target.

Theorem 29.1 (Modified-gravity classification). *Any proposed modification of Einstein form in the faithful metric projection is one of: same-scope collapse to Einstein form; skin; anchor change; hidden tensor load; higher-response curvature load; nonlocal continuation expansion; source-ledger extension; horizon certificate mechanism; richer-scope theory; or standing failure.*

Proof. If the modification satisfies SameScopeMetric₁, the normal-form and Einstein-combination theorems force Einstein form. If it changes no tensor content, it is skin. If it changes dimension, signature, connection, boundary, or locality, it changes anchors. If it adds fields or tensor terms, it imports tensor load. If it uses curvature products or derivatives, it is higher-response. If nonlocal, it expands continuation. If source-dependent, it extends the source ledger. If horizon-specific, it requires certificate closure. If none of these licenses apply, the modification fails standing. $\square$

Candidate	AASC classification
$G_{\mu\nu}$	same-scope survivor
$\Lambda g_{\mu\nu}$	homogeneous vacuum ledger
$R_{\mu\nu}$ alone	divergence failure
$Rg_{\mu\nu}$ alone	divergence failure
R^2	higher-response/effective tensor load
$R_{\mu\nu}R^{\mu\nu}$	higher-curvature tensor load
Weyl-squared	conformal/high-curvature tensor load
$f(R)$	selector function or scalar-response theory
scalar-tensor gravity	added scalar tensor witness
vector-tensor gravity	added vector tensor witness
torsion	new connection load
metric-affine nonmetricity	post-metric comparison/connection tensor load
extra dimensions	domain expansion
nonlocal gravity	continuation-envelope expansion
semiclassical gravity	mixed projection requiring record closure
EFT gravity	restricted-scope effective expansion
black-hole remnant	standing sink unless certified

Candidate	AASC classification
holographic rescue	global retention unless exterior certificate exists
radiation/leakage	possible leakage/certificate channel when independently licensed

Part XI

Main Theorem

30 Einstein dynamics as the unique minimal first-response metric projection

The theorem uses the local regular-stratum constraint role as the gravity-role input for the metric projection theorem. It does not assert global functional-analytic completion of ADM phase space, all singular strata, or all boundary-domain completions.

Theorem 30.1 (Main theorem). *Let a nondegenerate admissible physical interior instantiate the kernel $K = (\text{Adm}, \text{Stand}, \text{Ref}, \text{Irr})$, the fixed-domain consequence layer A^+ , quotient architecture, and the regular-stratum AER conditions sufficient to force the AASC gravity role. Let its faithful four-dimensional Lorentzian metric projection be $\Pi_g(A_G) = (M^4, g_{\mu\nu})$. Let $E_{\mu\nu}$ satisfy $\text{SameScopeMetric}_1(E_{\mu\nu})$ as a nontrivial general source-response candidate. Let $T_{\mu\nu}^{\text{closed}}$ be the closed same-scope source ledger fixed by boundary-trace role discipline and coherent trajectory closure. If homogeneous vacuum terms obey quotient discipline, then the unique minimal first-response same-scope metric field-equation form is*

$$G_{\mu\nu} + \Lambda g_{\mu\nu} = \kappa T_{\mu\nu}^{\text{closed}}.$$

Proof. The kernel and consequence layer fix target, standing, admissibility, reference, irreversibility, AMetric no-selector discipline, no same-act repair, and no hidden same-scope discriminator. Quotient architecture and the AER forcing theorem give the AASC gravity role: coisotropic standing, characteristic redescription, local first-class closure, and multiplier-type presentation. Faithful metric projection licenses (M, g, ∇) as geometric presentation, not primitive foundation. By the target-specific fixation ledger in Section 18, boundary-trace fixation fixes the metric, source, vacuum, and horizon/certificate roles as role-anchor equivalence classes without representative selection. Coherent trajectory dynamics gives closed source ledger conservation $\nabla^\mu T_{\mu\nu}^{\text{closed}} = 0$. The predicate

SameScopeMetric₁($E_{\mu\nu}$) applies to the geometric response side. By curvature-response exhaustion,

$$E_{\mu\nu} = aR_{\mu\nu} + bRg_{\mu\nu} + cg_{\mu\nu}.$$

Closed source compatibility requires $\nabla^\mu E_{\mu\nu} = 0$. The Einstein-combination theorem gives $b = -a/2$, hence

$$E_{\mu\nu} = aG_{\mu\nu} + cg_{\mu\nu}.$$

Since the theorem concerns the nontrivial general source-response branch, $a \neq 0$. Division by a is licensed. The $a = 0$ homogeneous branch has already been classified as vacuum-baseline ledger rather than general matter response. Coupling normalization yields

$$G_{\mu\nu} + \Lambda g_{\mu\nu} = \kappa T_{\mu\nu}^{\text{closed}}.$$

Vacuum quotient discipline blocks primitive zero-point interpretation of Λ . Any alternative inside the theorem target is skin, anchor motion, tensor-load import, richer scope, effective correction, nonlocal continuation expansion, vacuum scalar laundering, source-ledger incompleteness, or standing failure. High-curvature and horizon endpoint routes are handled by the route-closure corollary below rather than by the Einstein-form derivation itself.

□

Remark 30.2 (Scope of uniqueness). *Uniqueness is uniqueness of form within the declared four-dimensional Lorentzian metric projection envelope and up to normalization of the nonzero curvature coefficient, the source coupling κ , and the homogeneous ledger coefficient Λ . It is not uniqueness of all gravitational dynamics and not exclusion of declared richer-scope theories.*

Corollary 30.3 (High-curvature route closure). *High-curvature or horizon endpoint claims cannot function as hidden same-scope repairs of the minimal first-response metric equation. If they carry standing-bearing force, they must be declared as higher-response tensor load, effective-scope correction, lawful leakage/radiation channel, non-sink quotient, boundary certificate, or richer-scope theory. Otherwise they are skin, global-retention laundering, horizon standing sinks, or hidden repair.*

31 AASC-free reconstruction boundary

A conventional reconstruction is a different theorem with a different proof carrier. It would have to replace the kernel-forced metric-projection route by a metric-first theorem proving all of the following under explicit non-AASC assumptions and without hidden

imports: fixed target and carrier; lawful redescription; source-ledger closure; exclusion of hidden selectors; first-response same-scope exhaustion; vacuum quotient discipline; and high-curvature route closure. Such a reconstruction may corroborate, refine, or replace the AASC route only by identifying a failed named link or by proving the same target under fewer declared anchors without hidden imports.

Proposition 31.1 (Reconstruction route is separate). *A metric-first reconstruction is correctness-relevant to this manuscript only if it shows that one of the named AASC links fails or that it proves the same theorem under the same target with fewer declared anchors and without hidden selector, source, continuation, or scope import. Otherwise it is a parallel theorem, not a refutation.*

Proof. A parallel proof route can establish the same conclusion by different means, but it does not invalidate an existing proof unless it exposes an error in the existing proof's assumptions, inferences, or scope report. Since this manuscript declares its target and proof class, an AASC-free reconstruction is correctness-relevant only through a named link failure. $\square$

32 Correctness-relevant denial targets

A same-mode denial is correctness-relevant only if it identifies one of the following failures.

1. Target adequacy does not force the kernel.
2. The AMetric boundary does not block primitive metric backflow.
3. Physical standing does not require quotient architecture.
4. The AER interface does not force coisotropic standing, characteristic redescription, first-class closure, or multiplier form.
5. The AASC gravity role is not fixed by the physical interior.
6. Faithful metric projection improperly licenses or excludes metric structure.
7. SameScopeMetric₁ excludes a genuine same-scope first-response metric tensor.
8. The curvature-response normal form misses an admissible local symmetric rank-two tensor.
9. Divergence compatibility does not force the Einstein combination.
10. Closed source continuation is illicitly assumed rather than derived from coherent source trajectories.
11. Stress-energy extraction launders primitive mass-energy instead of extracting metric-facing source roles.

12. $\Lambda g_{\mu\nu}$ is misclassified as homogeneous quotient-stable ledger content.
13. High-curvature or horizon terms are same-scope first-response repairs rather than declared richer/effective/certificate routes.
14. Modified-gravity classification falsely rejects declared richer-scope theories.
15. The theorem overreports beyond minimal first-response four-dimensional Lorentzian metric projection.

33 Conclusion

AASC does not begin with a metric, curvature, scalar mass, or vacuum-energy number. It begins with admissibility, standing, reference, irreversibility, quotient identity, lawful redescription, and continuation. Sections above derive, in self-contained form, the gravity role as the constraint architecture required for the admissible physical interior. The present paper shows that when that role is faithfully projected into a minimal four-dimensional Lorentzian metric regime, and when trajectory coherence and boundary-trace role fixation are made explicit, the first-response metric curvature candidates are exhausted by Ricci response, scalar-trace response, and metric baseline:

$$E_{\mu\nu} = aR_{\mu\nu} + bRg_{\mu\nu} + cg_{\mu\nu}.$$

Closed source-ledger continuation fixes the Einstein combination:

$$E_{\mu\nu} = aG_{\mu\nu} + cg_{\mu\nu}.$$

The source side is the metric-facing tensor ledger of AASC deformation-response, field-carrier, particle, transport, interaction, and vacuum-baseline roles. The cosmological term is the quotient-stable homogeneous vacuum ledger, not primitive zero-point prediction. High-curvature and horizon regimes require declared tensor load, lawful leakage/radiation channel, independently licensed endpoint-leakage certificate, non-sink quotient, or richer scope.

Therefore

$$\boxed{G_{\mu\nu} + \Lambda g_{\mu\nu} = \kappa T_{\mu\nu}^{\text{closed}}}$$

is the unique minimal first-response same-scope metric projection of AASC gravity.

A Corpus support map

This appendix records provenance and full-source expansion roles. The main text contains the theorem chain needed to read the Einstein proof; the corpus papers provide expanded proofs, variants, and audit lineage.

Support cluster	Function in the Einstein projection theorem
Gravity as Structural Necessity	Full-source expansion of the in-manuscript AASC gravity-role derivation: coisotropic forcing, characteristic redescription, first-class closure, multiplier-type Hamiltonian rigidity, ADM realization, and projection discipline.
Metricity as Licensed Description	Supplies no primitive metricity, metric projection interface, L-metric boundary, metric tensor role, and no-backflow discipline.
Anchor, Tensor, and Skin	Supplies the fixed-domain object grammar: anchors, tensors, skin, no fourth role, non-generation, and role factorization.
Boundary-trace fixation	Supplies role-anchor fixation without representative selection for metric, source, vacuum, and certificate roles.
Bivalent trajectory theory	Supplies legal trajectory, coherent trajectory, failure witness, no deferred coherence, irrecoverability of failure, and no skin repair.
Mass, Inertia, and Persistence Compression Vacuum role and ZPE arcs	Supplies mass/inertia as deformation-response and transport-state response, not primitive scalar source. Supply non-null vacuum baseline, no primitive vacuum-energy assignment, homogeneous vacuum sector, quotient-descent obstruction, homogeneous shift, and renormalization-standing discipline.
Entanglement as boundary compatibility	Supplies nonmetric compatibility without communication or hidden transport.
Horizon instability arc	Supplies no standing sinks, certificate necessity, leakage-type route closure, remnant obstruction, and exterior reconstruction discipline.
UEAP	Supplies target fixation, carrier standing, alternative exhaustion, laundering exclusion, and report-preserving output for the theorem’s claim status.

The provenance expansions are used only at their stated type. AMetric non-selection is not a field equation. ATS is not a curvature proof. UEAP is not evidence for the theorem. Vacuum non-nullity is not a numerical value for Λ . Horizon closure is not a prediction of endpoint dynamics. The present paper obtains the Einstein field-equation form only after faithful four-dimensional Lorentzian metric projection and SameScopeMetric₁ exhaustion.

B Formal predicate SameScopeMetric₁($E_{\mu\nu}$)

The predicate SameScopeMetric₁($E_{\mu\nu}$) defines the minimal first-response same-scope metric target. It is intentionally narrow. It does not prohibit higher-curvature, metric-affine, scalar-tensor, nonlocal, or semiclassical theories; it classifies them outside the minimal first-response target unless they collapse to the linear form.

- (S1) Metric-internality: $E_{\mu\nu}$ is built only from (M^4, g, ∇) , where ∇ is the minimal metric-compatible connection.
- (S2) Tensoriality: $E_{\mu\nu}$ is tensorial under lawful redescription.
- (S3) Rank and symmetry: $E_{\mu\nu}$ is a symmetric covariant rank-two tensor.
- (S4) Locality: $E_{\mu\nu}$ is local in the metric projection.
- (S5) First-response order: $E_{\mu\nu}$ is linear in the first admitted curvature obstruction.
- (S6) No curvature products or derivatives: products, powers, arbitrary functions, and derivatives of curvature are higher-response or richer-scope unless declared.
- (S7) No extra fields, torsion, metric-affine nonmetricity, nonlocal operators, boundary/global data, representative-dependent terms, or hidden source content.
- (S8) Closed-source compatibility: $E_{\mu\nu}$ must be capable of matching a boundary-trace fixed closed source ledger.

In ATS terms, SameScopeMetric₁ enforces fixed anchors, declared tensor content, and no skin carrying force. A term outside SameScopeMetric₁ may be meaningful, but it is not an undeclared same-scope first-response metric term.

C Curvature-response exhaustion proof

Lemma C.1 (Tensoriality). *Connection coefficients alone are not tensorial under arbitrary redescription. A connection-only expression is skin or representative privilege unless completed into curvature. The first tensorial obstruction of compatible metric transport is curvature.*

Lemma C.2 (Rank type). *A closed source ledger extracted from metric response has symmetric rank-two type. A direct metric response must therefore be symmetric rank two. Scalars require metric return, higher-rank objects require contraction or extra tensor structure, and antisymmetric components require torsion, spin, or connection load.*

Lemma C.3 (First curvature slot). *The curvature of the minimal metric connection has Riemann type $R^\rho{}_{\sigma\mu\nu}$. A rank-two metric-internal first response is obtained by contraction to the Ricci tensor $R_{\mu\nu}$. Other metric-internal contractions reproduce Ricci/scalar information, vanish by curvature symmetries, or require additional structure.*

Trace-free Ricci is not a fourth first-response slot. In dimension n ,

$$R_{\mu\nu}^{TF} = R_{\mu\nu} - \frac{1}{n}Rg_{\mu\nu},$$

and in the four-dimensional projection,

$$R_{\mu\nu}^{TF} = R_{\mu\nu} - \frac{1}{4}Rg_{\mu\nu}.$$

It is already in the span of the Ricci and scalar-trace slots.

Lemma C.4 (Trace return). *The scalar trace R is not rank two. In the minimal metric projection the only available rank-two metric return is $Rg_{\mu\nu}$. Any other return uses auxiliary tensor or source structure.*

Lemma C.5 (Metric baseline). *The only zero-curvature same-scope symmetric rank-two metric baseline is $g_{\mu\nu}$, up to a projection-envelope coefficient. A nonconstant scalar multiplier requires extra field, boundary, or selector structure.*

Lemma C.6 (Orientation, parity, and topological terms). *A term requiring an orientation tensor, density, parity-odd contraction, Pontryagin/Euler density, or topological boundary datum is not a minimal same-scope first-response symmetric rank-two metric tensor unless the relevant orientation, density, topological, or boundary anchor is declared. If declared, it is boundary/global or richer-scope structure rather than an element of SameScopeMetric₁.*

Lemma C.7 (Exclusion). *Curvature products are higher-response terms. Curvature derivatives import higher continuation memory. Arbitrary functions of curvature import selectors and usually extra scalar response. Torsion changes the connection tensor ledger. Metric-affine nonmetricity changes the post-metric comparison/connection structure. Extra fields add tensor or source load. Nonlocal terms expand the continuation envelope. Boundary and global terms are anchors. Coordinate terms are skin. Horizon endpoint terms require certificate, leakage, non-sink quotient, or richer scope. No candidate outside $R_{\mu\nu}$, $Rg_{\mu\nu}$, and $g_{\mu\nu}$ remains a minimal same-scope first-response metric tensor.*

Candidate family	Mathematical status	AASC first-response status
Connection-only terms	Available as coefficients of a chosen connection presentation.	Not tensorial under re-description unless completed into curvature; skin or representative privilege.
Full Riemann tensor	First curvature obstruction with four indices.	Must be contracted or paired with extra structure; contraction yields Ricci/scalar slots.
Trace-free Ricci	Rank-two symmetric tensor.	Linear recombination of Ricci and scalar-trace slots; not independent.
Weyl tensor	Conformal curvature content.	Not a direct symmetric rank-two source-response tensor without differentiation, squaring, projection, or auxiliary structure.
Curvature products	Standard higher-curvature mathematical terms.	Higher-response/effective tensor load, not first-response.
Curvature derivatives	Valid in higher-derivative theories.	Higher continuation memory; outside minimal same-scope response.
Arbitrary functions such as $f(R)$	Legitimate richer-scope ansatz when declared.	Function selector and extra scalar response unless linearized.
Torsion	Independent connection structure.	Declared post-metric connection tensor load.
Metric-affine nonmetricity	Post-metric failure of metric preservation by connection.	Declared richer-scope comparison/connection load, distinct from AMetric boundary nonmetricity.
Scalar/vector/tensor additions	Common modified-gravity structures.	Added source/tensor witnesses.
Nonlocal curvature terms	Mathematically meaningful in nonlocal theories.	Continuation-envelope expansion.

Candidate family	Mathematical status	AASC first-response status
Topological or parity-odd terms	Boundary, orientation, or density-sensitive structures.	Require declared orientation/topological/boundary anchors.
Quantum expectation values	Semiclassical source candidates.	Mixed projection requiring state, record, renormalization, and source closure.
Horizon endpoint terms	Boundary/exterior reconstruction structures.	Certificate or richer-scope route; not hidden first-response repair.

Theorem C.8 (Normal form). *If $\text{SameScopeMetric}_1(E_{\mu\nu})$, then*

$$E_{\mu\nu} = aR_{\mu\nu} + bRg_{\mu\nu} + cg_{\mu\nu},$$

where a, b, c are projection-envelope coefficients. They are not *AMetric* defaults or boundary-generated numerical values; their dimensional interpretation requires source coupling, scale anchoring, or calibration.

Proof. Tensoriality excludes connection-only presentation. Rank type fixes symmetric rank-two target type. The first-curvature slot lemma yields $R_{\mu\nu}$. Trace return yields $Rg_{\mu\nu}$. Metric baseline yields $g_{\mu\nu}$. The orientation/topology and exclusion lemmas remove all other undeclared same-scope candidates. Linear combination of the surviving slots gives the normal form. $\square$

D Divergence calculation and coefficient discipline

Start with the normal form

$$E_{\mu\nu} = aR_{\mu\nu} + bRg_{\mu\nu} + cg_{\mu\nu}.$$

Closed source continuation requires $\nabla^\mu E_{\mu\nu} = 0$. Inside the licensed metric projection,

$$\nabla^\mu R_{\mu\nu} = \frac{1}{2}\nabla_\nu R, \quad \nabla^\mu g_{\mu\nu} = 0.$$

Therefore

$$\nabla^\mu E_{\mu\nu} = a\nabla^\mu R_{\mu\nu} + b\nabla^\mu(Rg_{\mu\nu}) + c\nabla^\mu g_{\mu\nu} = \left(\frac{a}{2} + b\right)\nabla_\nu R.$$

General continuation compatibility gives $b = -a/2$, so

$$E_{\mu\nu} = a \left(R_{\mu\nu} - \frac{1}{2} R g_{\mu\nu} \right) + c g_{\mu\nu} = a G_{\mu\nu} + c g_{\mu\nu}.$$

The constants a, b, c are not produced at the AMetric boundary. They are projection-envelope coefficients, and their dimensional interpretation enters only after source coupling and scale anchoring. Normalization gives

$$G_{\mu\nu} + \Lambda g_{\mu\nu} = \kappa T_{\mu\nu}^{\text{closed}}.$$

E Boundary-trace and coherent trajectory details

Boundary-trace fixation supplies role-anchor equivalence classes without selecting representatives:

$$\begin{aligned} [MetricRole]_{\equiv Role}, \quad [SourceRole]_{\equiv Role}, \quad [VacuumLedgerRole]_{\equiv Role}, \\ [HorizonCertificateRole]_{\equiv Role}. \end{aligned}$$

This prevents the metric, source, vacuum, and certificate roles from being primitive representatives or hidden selectors.

A coherent trajectory is a legal same-scope trajectory that preserves standing throughout its retained lineage. No deferred coherence, no skin repair, and irrecoverability of failure block post-hoc source repair. For a closed metric/source trajectory, coherence implies

$$\nabla^\mu T_{\mu\nu}^{\text{closed}} = 0.$$

If this condition fails, then the source ledger is incomplete, exchange with another ledger is undeclared, the system is open/effective, anomaly/failure has entered, scope changed, or hidden source transport has been imported.

In ordinary GR language this is the familiar covariant stress-energy conservation condition. In AASC it is not imported as primitive GR law; it is the metric projection of closed coherent source continuation.

F Source-ledger extraction

A closed AASC source ledger admits a metric-facing stress-energy tensor when the variation of source standing under metric projection can be represented by

$$\delta_g \mathcal{L}_{source}^{closed} = -\frac{1}{2} \sqrt{-g} T_{\mu\nu}^{closed} \delta g^{\mu\nu}.$$

Where an action formalism is used as extraction device,

$$T_{\mu\nu}^{closed} = -\frac{2}{\sqrt{-g}} \frac{\delta S_{source}^{closed}}{\delta g^{\mu\nu}}.$$

The formula is an extraction formalism, not an action-primitivity assumption.

The source sequence is

$$\Pi_D \rightarrow \mu_D(\Pi) \rightarrow \iota_D(\Pi) \rightarrow T_{\mu\nu}^{closed}.$$

Particle identity supplies the localized persistence class. Mass supplies the deformation-response profile. Inertia supplies the transport-facing restriction. The stress-energy tensor is the metric-facing closed ledger of those source roles, together with field-carrier, interaction, restoration, and vacuum-baseline contributions when declared. Primitive scalar mass and primitive energy ledgers do not by themselves carry source standing.

Source-side object	Metric-facing status
Localized particle persistence class	Supplies the identity target whose deformation response can enter the source ledger.
Mass-role profile $\mu_D(\Pi)$	Standing-preserving deformation-response profile; not primitive scalar substance.
Inertia-role profile $\iota_D(\Pi)$	Transport-state restriction of the deformation-response profile.
Field-carrier mediation	Carrier structure supporting localized persistence and excitation.
Interaction/restoration burden	Declared tensor content when it changes standing-bearing source continuation.
Vacuum baseline	Non-null no-particle baseline; contributes only through extraction ledger.
Boundary/effective source terms	Admissible only when declared as scope, boundary, or effective ledgers.

G Vacuum/ZPE quotient ledger

The vacuum role is the non-null no-particle baseline

$$\Omega_D(U) = (\Phi_U^0, \partial_U^0, T_U^0, Q_U^0, G_U^0, \mu_U^0, R_U^0).$$

No-particle is not nullity. Vacuum-energy scalars are not primitive source content before extraction ledger fixation.

In the homogeneous vacuum sector,

$$\langle T_{\mu\nu} \rangle_{0,R} = -\rho_R g_{\mu\nu}.$$

The standing-bearing combination is

$$\Lambda_{\text{eff}} = \Lambda_R + 8\pi G_R \rho_R,$$

not the split (Λ_R, ρ_R) by itself. A homogeneous shift

$$\rho_R \mapsto \rho_R + \delta\rho, \quad \Lambda_R \mapsto \Lambda_R - 8\pi G_R \delta\rho$$

preserves the effective metric-facing term. The Einstein-form theorem therefore licenses $\Lambda g_{\mu\nu}$ as quotient-stable homogeneous ledger content; it does not license a cutoff zero-point prediction for the numerical value of Λ .

H Entanglement and nonmetric compatibility

AMetric compatibility is non-transmissive. It is not communication, signaling, hidden transport, or repair. Entanglement illustrates the general rule: joint tensor compatibility may be fixed at admission before metric distance and time are admissibility-relevant; later metric representations can display spacelike-separated correlations without boundary-level transmission.

Thus

$$\text{BoundaryCompatibility} \neq \text{InteriorTransport},$$

$$\text{AMetricCompatibility} \neq \text{NonmetricCommunication}.$$

A horizon, nonlocal, or high-curvature model cannot preserve standing by appealing to nonmetric communication. If it claims standing transfer, it must supply exterior witness algebra, leakage/radiation channel, boundary-to-boundary certificate, non-sink quotient, or declared richer-scope tensor load.

I Horizon and high-curvature route closure

This module is not part of the derivation of the Einstein tensor. It closes the route by which high-curvature or horizon endpoint claims might otherwise function as hidden same-scope repairs.

A high-curvature or horizon response is admissible on the same AASC metric-projection scope only if it is one of: skin-equivalent presentation of exterior standing content, declared high-curvature tensor/effective load, lawful leakage/radiation certificate channel, boundary-to-boundary standing certificate, non-sink quotient reconstruction, or declared richer-scope theory. Otherwise it is hidden repair, global retention laundering, remnant compression, selector import, or scope drift.

Route	AASC status
Permanent horizon without certificate	Standing sink; inadmissible as same-scope preservation.
Remnant without exterior map	Compression obstruction; hidden storage does not preserve exterior standing.
Global unitarity/retention claim	Not exterior reconstruction without recoverability certificate.
Holographic rescue	Admissible only if it induces an exterior certificate in the declared scope.
Leakage/radiation/ endpoint-leakage channel	Possible certificate route when independently licensed; not predicted by this theorem.
Higher-curvature correction	Declared higher-response/effective tensor load; not hidden first-response repair.

J Modified-gravity taxonomy

Candidate	AASC classification
$G_{\mu\nu}$	Same-scope survivor after SameScopeMetric ₁ and divergence compatibility.
$\Lambda g_{\mu\nu}$	Homogeneous quotient-stable vacuum ledger.
$R_{\mu\nu}$ alone or $Rg_{\mu\nu}$ alone	Divergence failure unless combined into Einstein form.

Candidate	AASC classification
R^2 , $R_{\mu\nu}R^{\mu\nu}$, Weyl-squared $f(R)$	Higher-response/effective tensor load. Selector function and extra scalar response unless linearized to the first-response class.
Scalar-tensor/vector-tensor	Added tensor/source witness; richer scope.
Torsion	Post-metric connection tensor load.
AMetric nonmetricity	Foundational pre-metric boundary condition, not a modified-gravity term.
Metric-affine nonmetricity $Q_{\lambda\mu\nu}$	Post-metric connection/comparison tensor load.
Nonlocal gravity	Post-admission continuation-envelope expansion, not AMetric compatibility.
Semiclassical gravity	Mixed projection requiring state, record, and source closure.
Horizon endpoint theory	Exterior reconstruction/certificate problem.

Subscope	Added structure	Status relative to this theorem
Minimal first-response metric scope	Four-dimensional Lorentzian metric, Levi-Civita compatibility, closed source ledger, first curvature response.	The theorem target.
Higher-response metric scope	Curvature products, derivatives, or effective operators.	Declared effective/richer-scope extension.
Metric-affine scope	Independent connection, torsion, or post-metric nonmetricity.	Altered connection/comparison ledger.
Matter-enriched scope	Additional scalar, vector, spinor, or tensor source fields.	Added source/tensor witnesses.
Semiclassical scope	Quantum state, expectation value, renormalization and record ledger.	Mixed projection; requires state and record closure.
Horizon/exterior scope	Exterior reconstruction, trapped regions, remnant, or endpoint behavior.	Requires certificate, leakage, non-sink quotient, or richer scope.

Subscope	Added structure	Status relative to this theorem
Cosmological effective scope	Homogeneous residuals, effective fluids, dark-sector terms.	Requires extraction, calibration, and reportability ledger.
Quantum-gravity realization scope	Joint GR/QM architecture and boundary records.	Constraint-native realization problem, not primitive quantization.

K Scale and endpoint discipline

AASC does not treat dimensional numerical values as primitive outputs. A dimensional value has standing only relative to a scale anchor, comparison carrier, calibration ledger, or source ledger.

Quantity	Endpoint status
κ	Coupling-scale anchor; numerical value requires calibration.
Λ	Homogeneous effective vacuum/metric ledger; numerical value requires cosmological extraction and calibration.
Matter mass scales	Source/mass anchors; not generated by the Einstein-form theorem.
Dimensionless ratios	May stand once the relevant comparison carrier is fixed.
Dimension/signature	Projection anchors unless derived in a separate theorem.

This is not a weakness of the theorem. The theorem derives form and role typing. Absolute scale is a different endpoint type.

Ordinary GR expression	AASC role typing
Diffeomorphism covariance	Lawful redescription / quotient identity.

Ordinary GR expres- sion	AASC role typing
Metric compatibility	Minimal metric-continuation preservation in the faithful projection.
Bianchi identity	Internal curvature identity of the admitted metric grammar.
Covariant source conserva- tion	Metric projection of closed coherent source continua- tion.
Stress-energy tensor	Metric-facing extraction of the closed source ledger.
Cosmological constant term	Homogeneous quotient-stable vacuum/metric ledger.
Higher-curvature correc- tions	Higher-response or effective tensor load when de- clared.
Matter action	Optional extraction formalism for already admitted source roles; not a primitive proof source.

L Standalone foundation theorem ladder

This appendix records the foundation chain used in the main text. The full corpus gives longer proofs and variants; the following ladder is sufficient for the Einstein theorem.

Foundation link	Einstein-theorem function
Target adequacy $\Rightarrow K$	A determinate metric-response theorem act al- ready requires admissibility, standing, reference, and irreversibility.
$K \Rightarrow A^+$	Supplies AMetric no-selector boundary, no same- act repair, no second same-scope classifier, report preservation, and no hidden carrier transfer.
Physical standing $\Rightarrow$ quotient architecture	Blocks representative privilege and fixes lawful redescription as quotient identity.
Quotient/AER interface	Provides the regular local Hamiltonian observable-evaluation regime without assuming Einstein equations.
AER forcing	Forces coisotropic standing, characteristic re- description, local first-class closure, and multi- plier form.

Foundation link	Einstein-theorem function
AASC gravity necessity	Identifies the resulting GR-type constraint architecture as gravity in the AASC sense.
Faithful metric projection	Represents the gravity role in local Lorentzian metric language without metric backflow.
SameScopeMetric ₁ exhaustion	Performs the new Einstein-side proof: no undeclared first-response same-scope metric tensor survives outside $R_{\mu\nu}$, $Rg_{\mu\nu}$, and $g_{\mu\nu}$.
Closed source continuation	Forces divergence-compatible geometric response and hence the Einstein combination.

M Adversarial audit matrix

Objection	Standing status	Response and named link
AASC is optional vocabulary.	No, unless a role failure is shown.	A determinate same-scope metric-response act already fixes target, standing, admissibility, reference, irreversibility, continuation, and reportability. Rejecting the names does not remove the roles. Link: kernel.
The theorem assumes GR.	Standing only if GR equations are found among premises.	The in-manuscript gravity theorem supplies constraint architecture, not Einstein dynamics. The field equation follows only after metric projection and SameScopeMetric ₁ exhaustion. Link: AER gravity role.
This is just Lovelock.	Comparator only.	Lovelock-style theorems begin after metric assumptions are admitted. AASC first licenses the metric projection and source/continuation standing. Link: SameScopeMetric ₁ .
The stress tensor is assumed.	Standing if primitive source content is used.	The stress tensor is the metric-facing extraction of a closed source ledger; primitive mass-energy substance is not a premise. Link: source ledger.

Objection	Standing status	Response and named link
Conservation is assumed.	Standing if imported from GR.	Closed source divergence follows from coherent same-scope source continuation. It is not imported as a GR postulate. Link: closed source continuation.
The cosmological constant is arbitrary.	Standing only against numerical overclaim.	The theorem licenses the homogeneous quotient-stable ledger form $\Lambda g_{\mu\nu}$; numerical value requires calibration and is not claimed. Link: vacuum quotienting.
Modified gravity falsifies the theorem.	Standing if a term satisfies SameScopeMetric ₁ without Einstein collapse.	Declared richer-scope/effective theories are not denied. They are outside the minimal first-response same-scope target unless they collapse to Einstein form. Link: modified-gravity taxonomy.
High curvature invalidates minimality.	Standing if hidden same-scope first-response repair is shown.	High-curvature terms require declared higher-response/effective tensor load, certificate, leakage, non-sink quotient, or richer scope. Link: horizon route closure.
Nonmetric compatibility allows hidden transport.	Standing if boundary compatibility can carry same-scope transport.	AMetric compatibility is not communication. Any transport claim must be declared as tensor load, ledger, certificate, or scope change. Link: nonmetric compatibility.
The constants are not predicted.	Not an objection to this target.	Absolute dimensional values require scale/calibration anchors. The theorem reports form standing, not numerical scale generation. Link: scale endpoint discipline.
This is philosophical, not mathematical.	No, unless a proof link is attacked.	The manuscript states definitions, theorems, normal forms, divergence calculation, and classification cases. Style classification is not a correctness criterion. Link: proof-class lock.

Objection	Standing status	Response and named link
A metric-first derivation is required.	Proof-class mismatch unless a link fails.	A metric-first derivation is a parallel reconstruction route, not a prerequisite for this proof class. Link: AASC-free reconstruction boundary.

N ATS/UEAP audit and objections

ATS and UEAP are governance/audit layers. ATS governs the standing-bearing status of objects inside the fixed domain. UEAP governs the reportable status of the theorem claim. Neither is evidence for the curvature-response theorem; the UEAP audit records target, carrier, scope, blockers, and reportability rather than proving the theorem.

Object	ATS role	Reason
AMetric boundary	Anchor	Pre-admissible non-selection condition.
Four-dimensional Lorentzian projection	Anchor	Projection envelope for theorem target.
Metric compatibility	Anchor in the minimal projection	Defines Levi-Civita comparison/transport envelope.
$R_{\mu\nu}, Rg_{\mu\nu}, G_{\mu\nu}$	Tensor	Standing-bearing curvature-response content.
$T_{\mu\nu}^{\text{closed}}$	Tensor	Closed metric-facing source ledger.
$\Lambda g_{\mu\nu}$	Tensor	Homogeneous vacuum ledger.
Coordinates, frames, notation	Skin	Presentation only.
External metric-uniqueness comparison	Skin/comparator	External orientation only; no proof authority.

UEAP coordinate	Audit value
Claim	Einstein dynamics is the unique minimal first-response same-scope metric projection of AASC gravity.

UEAP coordinate	Audit value
Target	Metric field-equation form inside the declared first-response projection envelope.
Carrier	AASC gravity role, ATS role typing, boundary-trace/coherent-trajectory support, SameScopeMetric ₁ , curvature-response exhaustion, stress-energy extraction, and vacuum quotienting.
Final status	Legitimate within declared minimal first-response metric-projection scope.
Non-reportable promotions	All gravitational dynamics, numerical constants, all matter dynamics, all modified gravity false, complete quantum gravity, complete horizon endpoint dynamics.

Laundering route	Blocked promotion	Correct report
Comparator laundering	Treat an external metric classification as proof authority.	Comparator only; AASC proof uses SameScopeMetric ₁ .
Zero-point-energy laundering	Treat cutoff vacuum sums as prediction of Λ .	Only quotient-stable homogeneous ledger enters.
Scale laundering	Treat κ or Λ as boundary-generated numbers.	Dimensional values require scale/calibration anchors.
All-gravity over-reporting	Report minimal first-response result as all gravitational dynamics.	Legitimate only in declared projection scope.
Horizon repair laundering	Use high curvature or horizons as hidden same-scope repair.	Requires certificate, leakage, non-sink quotient, or richer scope.
Source-substance laundering	Treat stress-energy as primitive mass-energy substance.	Source tensor is metric-facing extraction from standing source roles.
Nonmetric-communication laundering	Treat AMetric compatibility as transport.	Boundary compatibility is non-transmissive.

Requested report	Carrier required	Status in this paper
Minimal first-response Einstein form	AASC gravity role, faithful metric projection, SameScopeMetric ₁ , closed source ledger.	Certified by main theorem.
All gravitational dynamics	Global regime classification, all richer scopes, all boundary completions.	Not reported.
Numerical value of κ	Scale/coupling calibration anchor.	Out of target.
Numerical value of Λ	Cosmological extraction and calibration ledger.	Out of target.
Full matter dynamics	Matter-sector Lagrangians and interaction carriers.	Not reported.
Equivalence-principle completion	Dedicated inertial/gravitational role-identification theorem.	Prepared, not claimed.
Quantum-gravity completion	Constraint-native joint GR/QM realization certificate.	Not reported.
High-curvature endpoint dynamics	Exterior certificate, leakage, non-sink quotient, or richer-scope model.	Route-closure only.

Proof step	Carrier	Failure blocked
AMetric-to-metric order	AMetric boundary and metric licensing.	Primitive metric backflow.
AASC gravity role	In-manuscript derivation of GR-type constraint architecture, with corpus expansion.	Treating curvature as primitive gravity.
ATS role typing	Anchor/tensor/skin decomposition.	Fourth-role and skin-load import.
Boundary-trace fixation	Role-anchor equivalence without representative selection.	Canonical representative or hidden selector.
Coherent trajectory closure	Legal/coherent source lineage.	Deferred coherence and source repair.

Proof step	Carrier	Failure blocked
SameScopeMetric ₁ predicate	Minimal first-response metric target.	Hidden higher-response scope drift.
Curvature normal form	Six-lemma exhaustion.	Unclassified geometric alternatives.
Divergence compatibil- ity	Closed source continuation plus curvature identity.	Source/geometric mis- match.
Stress-energy extraction	Metric response of closed source ledger.	Primitive source- substance import.
Vacuum quotienting	Homogeneous effective ledger.	Zero-point numerical overclaim.
Nonmetric compatibil- ity closure	Boundary-level compatibility discipline.	Hidden nonmetric com- munication.
Horizon route closure	Certificate/leakage/non-sink taxonomy.	High-curvature hidden repair.
Scale endpoint disci- pline	Calibration and comparison anchors.	Boundary-generated constants.

The theorem assumes GR. The in-manuscript gravity-role result is constraint architecture, not the Einstein equation. The field equation follows only after faithful metric projection and SameScopeMetric₁ exhaustion.

The theorem is just Lovelock in different language. Metric-uniqueness classifications operate after metric assumptions are admitted. The AASC proof asks how metric response obtains standing from AMetric non-selection and source/trajectory discipline; the comparator is not a premise.

The theorem defines away modified gravity. The theorem defines the minimal first-response same-scope class. Modified theories are typed as higher-response, effective, richer-scope, or boundary/certificate regimes when declared.

The stress tensor is assumed. The tensor type is extracted from metric-facing source response. Its content is AASC deformation-response, field-carrier mediation, particle persistence, transport, interaction, and vacuum baseline.

The cosmological term is arbitrary. The term is the quotient-stable homogeneous vacuum ledger. Its numerical value requires calibration and scale anchoring.

High curvature invalidates the result. High-curvature corrections may be admissible as declared higher-response/effective load. They do not function as hidden same-scope first-response alternatives.

Nonmetric compatibility allows hidden transport. A Metric compatibility is not communication. Any post-metric transport or recovery requires a declared ledger or certificate.

The constants are not predicted. Absolute dimensional constants require scale anchors. The theorem derives the form and role typing.

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